

Network and Information Security

Sem VI



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Course Code	Name of the Course			
116U01C602	Network & Information Security			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	--	20	30	50
				100

Course prerequisites:

Basics of Operating System and Computer Network.

Course Objectives:

1. To understand fundamental principles of information security.
- To gain knowledge on malicious & non-malicious programme errors & apply counter- measures.
3. To understand the various web attacks and its preventive mechanisms.
4. To apply different techniques to secure data in transit across data networks.
5. To study security laws and analyse the ethical issues.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1: Understand various security goals, threats, vulnerabilities and controls.
- CO2: Apply various cryptographic algorithms for software security.
- CO3: Identify, analyse and mitigate web attacks.
- CO4: Apply network security mechanisms for threat management.
- CO5: Understand investigation process, legal and ethical issues in security.

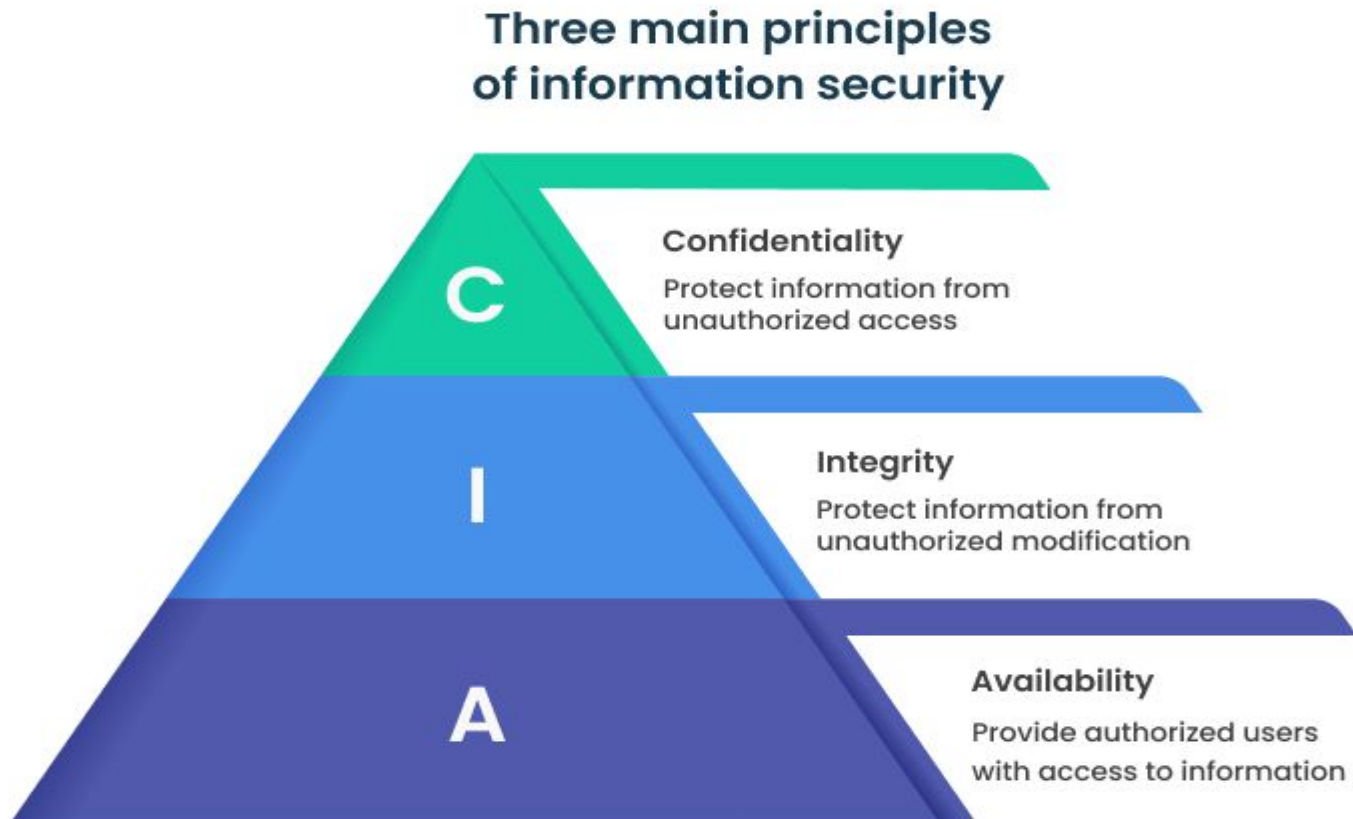
What is Information Security

- Information security is the practice of protecting information by mitigating information risks.
- It involves the protection of information systems and the information processed, stored, and transmitted by these systems from unauthorized access, use, disclosure, disruption, modification, or destruction.
- This includes the protection of personal information, financial information, and sensitive or confidential information stored in both digital and physical forms.
- Effective information security requires a comprehensive and multi-disciplinary approach, involving people, processes, and technology.

What is InfoSec?

- Information Security is basically the practice of preventing unauthorized access, use, disclosure, disruption, modification, inspection, recording, or destruction of information.
- Information can be a physical or electronic one. Information can be anything like Your details or we can say your profile on social media, your data on your mobile phone, your biometrics, etc.

3 Principles of Information Security?



- **Confidentiality**: Keeping data secret from unauthorized eyes (e.g., encryption, access controls).
- **Integrity**: Ensuring data is accurate and hasn't been altered (e.g., digital signatures, change management).
- **Availability**: Making sure authorized users can access data when needed (e.g., backups, disaster recovery)

Components & Practices

- People: Trained employees, security awareness.
- Processes: Policies, procedures, risk management, incident response.
- Technology: Firewalls, antivirus, encryption, cloud security, endpoint security.

Threat

- A threat is a possible security violation that may exploit a system's weakness. It can originate from accidents, natural disasters, human error, or intentional actions. The main threat categories include interruption, interception, fabrication, and modification.

- **Unstructured Threats:** Caused by inexperienced individuals using simple tools like shell scripts or password crackers.
- **Structured Threats:** Organized, targeted attempts by skilled and motivated hackers to breach a specific network or company.
- **External Threats:** Originating from outside the organization. Attackers gain unauthorized access via the internet or remote access points.
- **Internal Threats:** Caused by users who already have authorized access, whether through accounts or physical access.

Types of Attack

- **Active Attack:** The attacker alters, damages, or takes control of the system (e.g., malware injection, data theft, server crashes). Like someone breaking into your house and tampering with things.
- **Passive Attack:** The attacker silently monitors or steals information without making changes. Like someone secretly reading your messages.

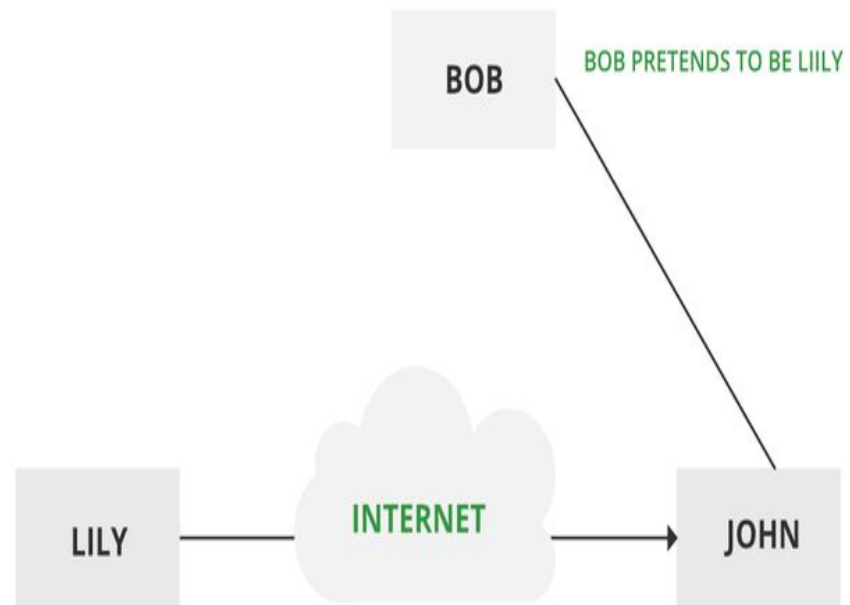
Types of active attacks

- Masquerade Attack
- Modification of Messages
- Repudiation
- Replay Attack
- Denial of Service(DoS) Attack

Masquerade Attack

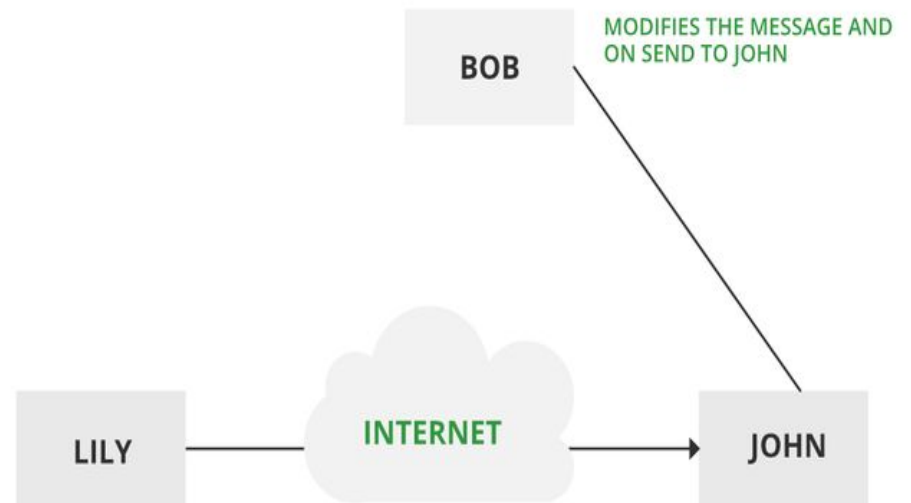
Masquerade attacks are considered one type of cyber attack in which the attacker disguises himself to pose as some other person and accesses systems or data.

- **Username and Password Masquerade**
- **IP address masquerade**
- **Website masquerade**
- **Email masquerade**



Modification of Messages

- This is when someone changes parts of a message without permission, or mixes up the order of messages, to cause trouble. Imagine someone secretly changing a letter you sent, making it say something different. This kind of attack breaks the trust in the information being sent. For example, a message meaning “Allow JOHN to read confidential file X” is modified as “Allow Smith to read confidential file X”.



Repudiation

- Repudiation attacks are a type of cyber attack wherein some person does something damaging online, such as a financial transaction or sends a message one does not want to send, then denies having done it. Such attacks can seriously hinder the ability to trace down the origin of the attack or to identify who is responsible for a given action, making it tricky to hold responsible the right person.

Types:

- **Message repudiation attacks**
- **Transaction repudiation attacks**
- **Data repudiation attacks**

Replay Attack

- It is a passive capturing of a message with an objective to transmit it for the production of an authorized effect.
- Thus, in this type of attack, the main objective of an attacker is saving a copy of the data that was originally present on that particular network and later on uses it for personal uses.
- Once the data gets corrupted or leaked it becomes an insecure and unsafe tool for its users.

Denial of Service(DoS) Attack

- Denial of Service (DoS) is a form of cybersecurity attack that involves denying the intended users of the system or network access by flooding traffic or requests. In this DoS attack, the attacker floods a target system or network with traffic or requests in order to consume the available resources such as bandwidth, CPU cycles, or memory and prevent legitimate users from accessing them.

Types of DoS attacks:

- **Flood attacks:** Here, an attacker sends such a large number of packets or requests to a system or network that it cannot handle them all and the system gets crashed.
- **Amplification attacks:** In this category, the attacker increases the power of an attack by utilizing another system or network to increase traffic then directs it all into the target to boost the strength of the attack.

Threat	Attack
A possible danger or risk.	Actual harmful action.
Can be intentional or unintentional.	Always intentional.
May or may not be malicious.	Always malicious.
May or may not cause damage.	Aims to cause damage.
Harder to detect.	Easier to detect.
Can be prevented by controlling vulnerabilities.	Cannot be stopped by vulnerability control alone.
May come from internal, external, physical, or non-physical sources.	Comes from malware, phishing, botnets, ransomware, DoS, breaches, etc.

Vulnerabilities

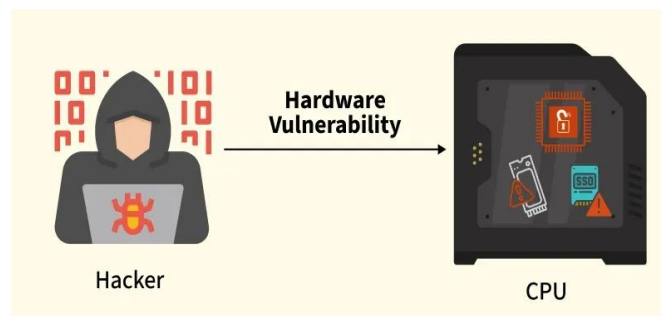
- Vulnerabilities are weaknesses in a system that give threats the opportunity to compromise an individual's or an organisation's assets.
- As the attackers are leveling up their attacking mechanisms, the number of vulnerable assets is increasing too.
- These could include the code, human weaknesses, unpatched software, improper authentication mechanisms, etc.

Types of Vulnerabilities

- **Hardware Vulnerability**
- **Software Vulnerability**
- **Network Vulnerability**
- **Procedural Vulnerability**
- **Human Vulnerability**

Hardware Vulnerability

- Weaknesses or flaws in physical devices (like computers or routers) that hackers can exploit to gain unauthorized access or cause damage.
- **Physical Attacks:** Hardware devices like servers, laptops, or smartphones are susceptible to physical attacks. Attackers may gain access to critical systems by stealing or tampering with hardware.
- **Firmware Vulnerabilities:** The software that runs on hardware, known as firmware, can have vulnerabilities. Flaws in firmware can lead to persistent attacks, as they are not always detected or patched as frequently as software



Software Vulnerability

Causes of Hardware Vulnerability

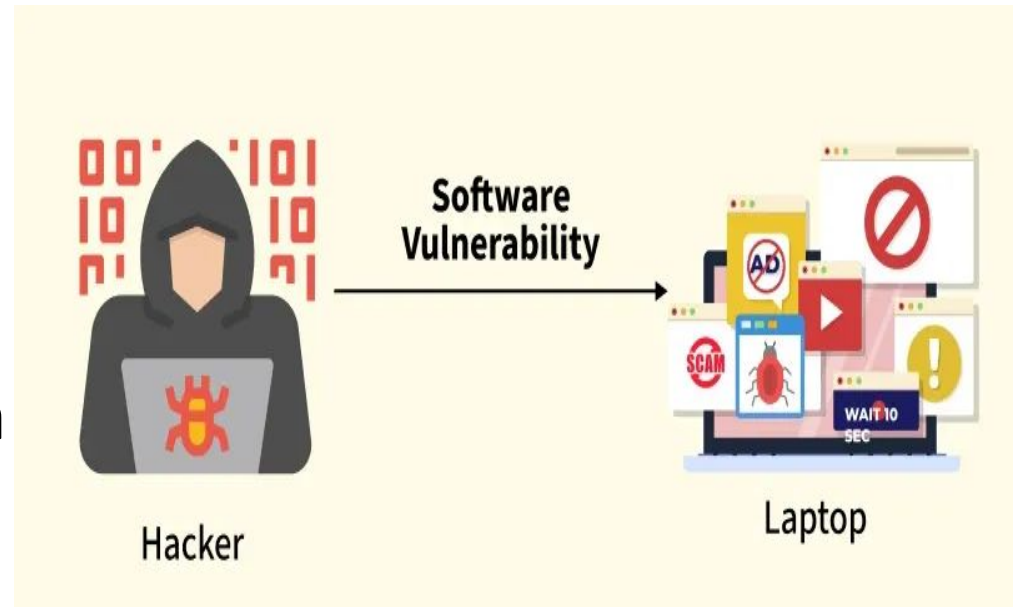
- Old version of systems or devices
- Unprotected storage
- Unencrypted devices, etc.

How to Prevent Hardware Vulnerability:

- **Encrypt Sensitive Data:** Ensure that all hardware, such as laptops and smartphones, is encrypted to protect data from unauthorized access in case of theft.
- **Use Strong Authentication:** Implement biometric or multi-factor authentication on devices to enhance security.
- **Secure Devices Physically:** Lock hardware devices in secure locations to prevent physical tampering or theft.
- **Keep Hardware Up-to-Date:** Regularly update hardware firmware to patch vulnerabilities and ensure the latest security measures are in place.

Software Vulnerability:

- Flaws or bugs in software (such as apps or operating systems) that can be used by hackers to compromise the system, often due to coding mistakes or outdated software.
- **Unpatched Software**
- **Buffer Overflow**
- **Insecure Code**
- **Weak Authentication**



Causes of Software Vulnerabilities

- Lack of input validation
- Unverified uploads
- Cross-site scripting
- Unencrypted data, etc.

How to Prevent Software Vulnerability:

- **Regular Patching and Updates:** Apply software patches and updates as soon as they are released. Many vulnerabilities arise due to outdated software.
- **Secure Coding Practices:** Developers should use secure coding practices to avoid common software vulnerabilities such as SQL injection and buffer overflows.
- **Use Antivirus Software:** Install and regularly update antivirus and anti-malware software to detect and remove malicious code.
- **Apply Proper Input Validation:** Ensure that input validation checks are in place to prevent attacks like cross-site scripting (XSS) and SQL injection.

Network Vulnerability

- A network vulnerability is a weakness or flaw in the design, implementation, or configuration of a computer network that attackers can exploit to gain unauthorized access, steal data, or disrupt services. These vulnerabilities can exist in hardware (routers, switches), software (servers, protocols), or network configurations.
- **Unsecured Wireless Networks:** Wi-Fi networks that are not properly secured with strong passwords or encryption can be easily accessed by attackers. Once inside the network, an attacker can intercept communications, gain access to devices, or launch attacks against connected systems.
- **Open Ports:** Unnecessary or open ports on a device can serve as gateways for attackers to exploit. Proper configuration of firewalls is essential to ensure only necessary ports are open and accessible.
- **Man-in-the-Middle (MITM) Attacks:** In MITM attacks, attackers intercept and alter the communication between two parties. If sensitive information such as login credentials or financial data is transmitted unencrypted, it can be captured and misused.

Causes of Network Vulnerability

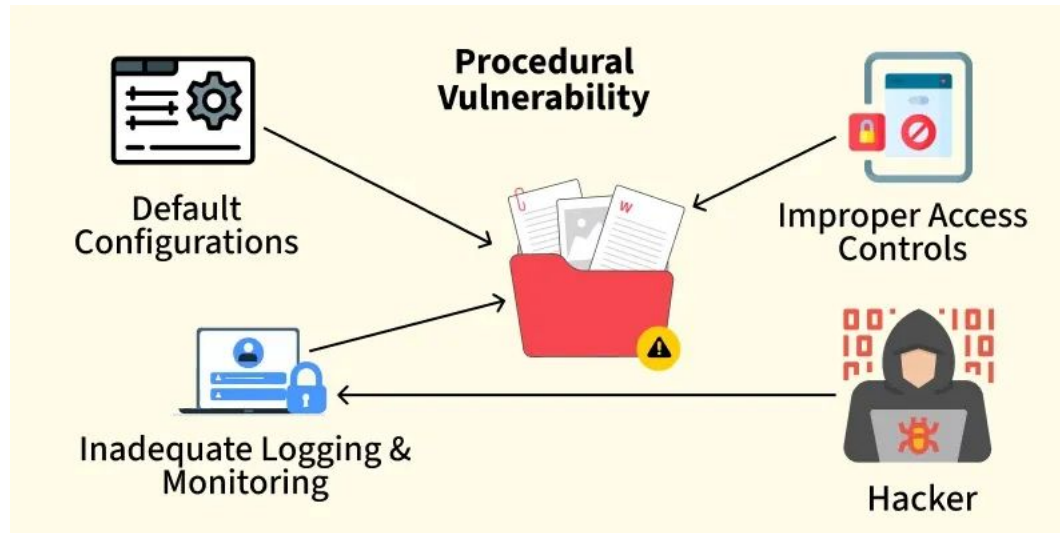
- Unprotected communication
- Malware or malicious software (e.g.:Viruses, Keyloggers, Worms, etc)
- Social engineering attacks
- Misconfigured firewalls

How to Prevent Network Vulnerability:

- **Use Strong Encryption:** Ensure that sensitive data is encrypted during transmission over the network using protocols like SSL/TLS.
- **Secure Wi-Fi Networks:** Use strong passwords and encryption (WPA3) on wireless networks to prevent unauthorized access.
- **Firewalls and IDS/IPS:** Implement firewalls to block unauthorized network traffic and use Intrusion Detection Systems (IDS) or Intrusion Prevention Systems (IPS) to monitor and protect network traffic.
- **Limit Open Ports:** Close unnecessary open ports on devices to reduce attack surfaces. Regularly audit the network for exposed ports.

Procedural Vulnerability

- Weaknesses in the processes or rules organizations follow, like using default passwords or failing to monitor activities, which can allow attackers to bypass security.



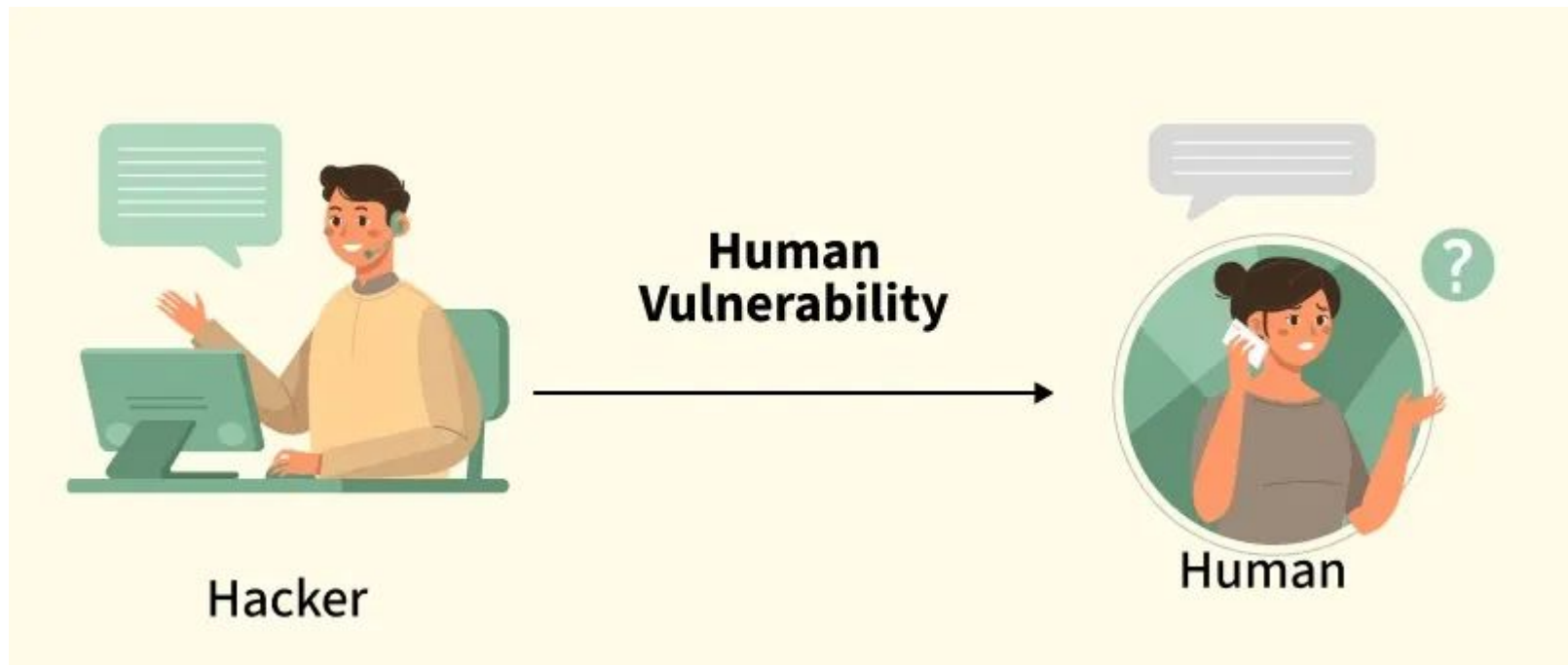
- **Default Configurations:** Many devices, applications, and systems come with default settings that are not optimized for security. Leaving these defaults unchanged, such as using default administrator passwords, can provide attackers with an easy way into the system.
- **Improper Access Controls:** Misconfiguring access control settings, such as leaving sensitive data accessible to unauthorized users, can expose systems to exploitation. Role-based access control (RBAC) and least privilege access are essential to minimize this risk.
- **Inadequate Logging and Monitoring:** Failure to implement adequate logging and monitoring can prevent organizations from detecting unauthorized access or malicious activities in a timely manner. A lack of monitoring can allow an attacker to maintain persistent access without detection.

How to Prevent Procedural Vulnerability:

- **Change Default Configurations:** Always change default configurations on devices, software, and network devices to ensure stronger security.
- **Role-Based Access Control (RBAC):** Implement RBAC to ensure that employees have access only to the information they need for their job.
- **Regular Monitoring and Logging:** Establish robust logging and monitoring practices to detect and respond to unusual activity promptly.
- **Use the Principle of Least Privilege (PoLP):** Only grant users the minimum level of access necessary for them to perform their duties.

Human Vulnerabilities

- Security risks caused by human behavior, such as falling for phishing attacks, using weak passwords, or not being aware of security threats, making it easier for hackers to exploit the system.

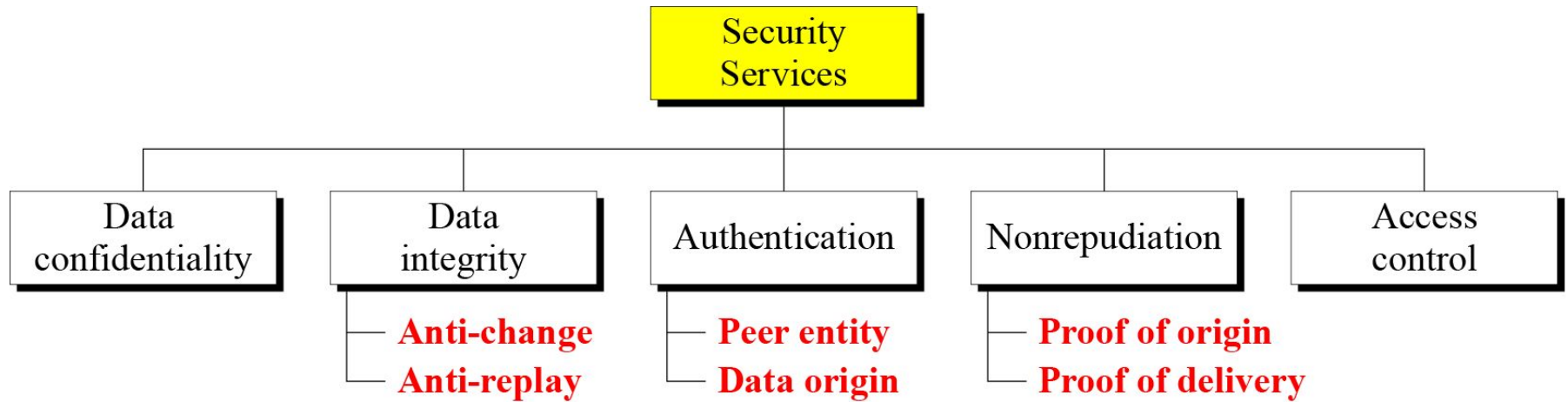


- **Social Engineering**: Human behavior is often the weakest link in cyber security. Attackers use social engineering tactics to manipulate individuals into disclosing confidential information or performing actions that compromise security. Phishing, baiting, and pretexting are common social engineering methods.
- **Negligence**: Employees or users may inadvertently introduce vulnerabilities through negligence, such as using weak passwords, sharing login credentials, or failing to lock their devices when not in use. This makes them easy targets for attackers.
- **Lack of Security Awareness**: A lack of training and awareness about cyber security best practices can leave individuals and organizations vulnerable to attacks. Users may fail to recognize phishing emails or may click on malicious links without thinking.

How to Prevent Human Vulnerability:

- **Security Awareness Training:** Conduct regular cyber security training for employees to help them identify phishing attacks, social engineering tactics, and other malicious activities.
- **Use Strong Passwords and Multi-Factor Authentication (MFA):** Encourage the use of complex passwords and MFA to secure user accounts.
- **Regular Security Audits:** Perform regular audits of employee behavior and access permissions to ensure compliance with security policies.
- **Be Cautious of Suspicious Emails and Links:** Train users to avoid clicking on unknown links, opening suspicious email attachments, or entering sensitive information on untrusted websites.

Figure 1.3 Security services



Security Services

Security services are mechanisms designed to **protect data and communication systems** from threats such as unauthorized access, modification, or misuse.

Data Confidentiality

Purpose: Ensures that information is **not disclosed to unauthorized users**.

How: Encryption, secure communication channels.

Example: Only the intended receiver can read an encrypted email.

Data Integrity

Purpose: Ensures that data is **not altered** during storage or transmission.

Sub-services:

Anti-change: Prevents unauthorized modification of data.

Anti-replay: Prevents attackers from capturing and re-sending old messages.

Example: Hash functions and Message Authentication Codes (MACs).

Authentication

Purpose: Confirms the **identity of entities** involved in communication.

Sub-services:

Peer entity authentication: Verifies the identity of the communicating parties (e.g., user login).

Data origin authentication: Confirms that data comes from a legitimate source.

Example: Passwords, biometrics, digital certificates.

Authentication

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Example: Passwords, biometrics, digital certificates.

Nonrepudiation

Purpose: Prevents a sender or receiver from **denying an action** they performed.

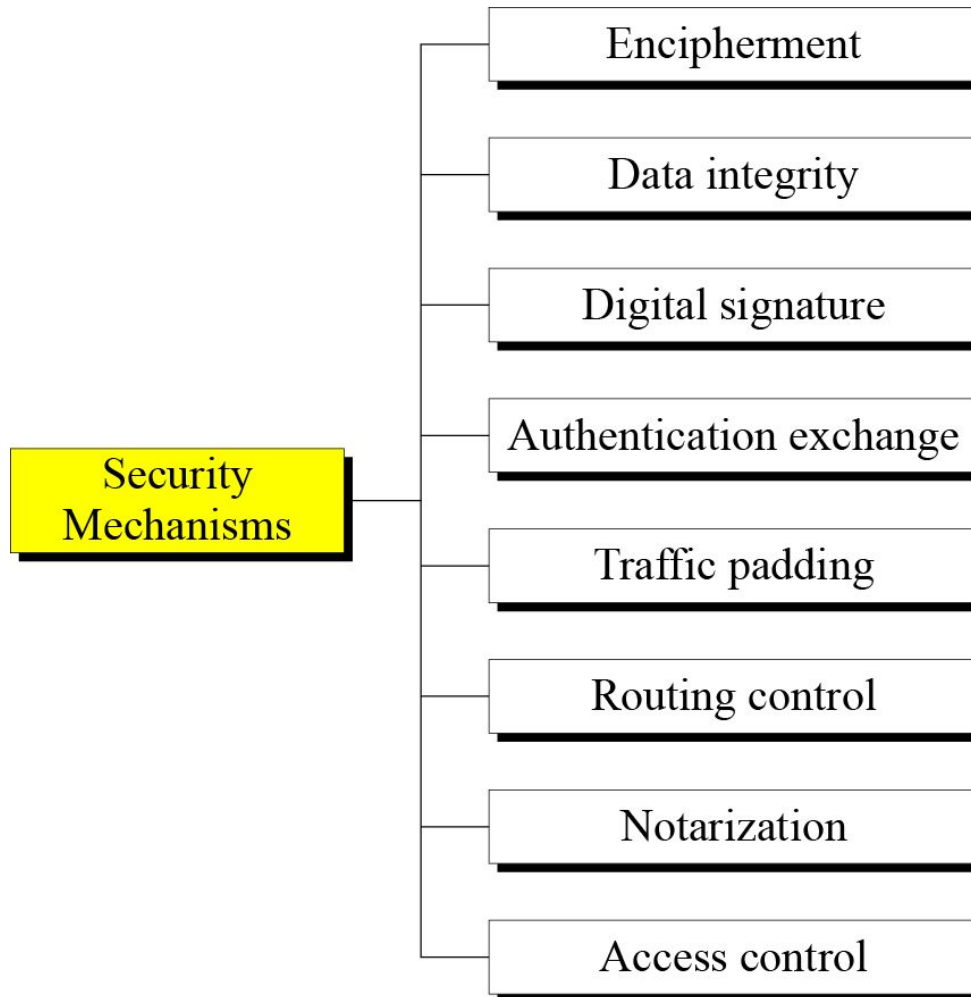
Sub-services:

Proof of origin: Sender cannot deny sending the message.

Proof of delivery: Receiver cannot deny receiving the message.

Example: Digital signatures in legal or financial transactions.

Figure 1.4 Security mechanisms



Encipherment : This security mechanism deals with hiding and covering of data which helps data to become confidential. It is achieved by applying mathematical calculations or algorithms which reconstruct information into not readable form. It is achieved by two famous techniques named [Cryptography](#) and Encipherment. Level of data encryption is dependent on the algorithm used for encipherment.

Access Control : This mechanism is used to stop unattended access to data which you are sending. It can be achieved by various techniques such as applying passwords, using [firewall](#), or just by adding PIN to data.

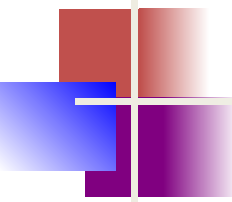
Notarization : This security mechanism involves use of trusted third party in communication. It acts as mediator between sender and receiver so that if any chance of conflict is reduced. This mediator keeps record of requests made by sender to receiver for later denied.

Data Integrity : This security mechanism is used by appending value to data to which is created by data itself. It is similar to sending packet of information known to both sending and receiving parties and checked before and after data is received. When this packet or data which is appended is checked and is the same while sending and receiving data integrity is maintained.

Authentication Exchange : This security mechanism deals with identity to be known in communication. This is achieved at the [TCP/IP layer](#) where two-way handshaking mechanism is used to ensure data is sent or not

Bit Stuffing : This security mechanism is used to add some extra bits into data which is being transmitted. It helps data to be checked at the receiving end and is achieved by [Even parity or Odd Parity](#).

Digital Signature : This security mechanism is achieved by adding digital data that is not visible to eyes. It is form of [electronic signature](#) which is added by sender which is checked by receiver electronically. This mechanism is used to preserve data which is not more confidential but sender's identity is to be notified.



1.3.3 Relation between Services and Mechanisms

Table 1.2 Relation between security services and mechanisms

<i>Security Service</i>	<i>Security Mechanism</i>
Data confidentiality	Encipherment and routing control
Data integrity	Encipherment, digital signature, data integrity
Authentication	Encipherment, digital signature, authentication exchanges
Nonrepudiation	Digital signature, data integrity, and notarization
Access control	Access control mechanism

Mechanisms discussed in the previous sections are only theoretical recipes to implement security. The actual implementation of security goals needs some techniques. Two techniques are prevalent today: cryptography and steganography.

Topics discussed in this section:

1.4.1 Cryptography

1.4.2 Steganography

Cryptography, a word with Greek origins, means “**secret writing.**” However, we use the term to refer to the science and art of transforming messages to make them secure and immune to attacks.

The word steganography, with origin in Greek, means “covered writing,” in contrast with cryptography, which means “secret writing.”

Example: covering data with text

This book is mostly about cryptography, not steganography.

□	□□□	□	□	□	□□□
0	1 0	0	0		0 1

Example: using dictionary

A	friend	called	a	doctor.
0	10010	0001	0	01001

Example: covering data under color image

0101001 <u>1</u>	10111110 <u>0</u>	0101010 <u>1</u>
0101111 <u>0</u>	10111110 <u>0</u>	0110010 <u>1</u>
0111111 <u>0</u>	0100101 <u>0</u>	0001010 <u>1</u>

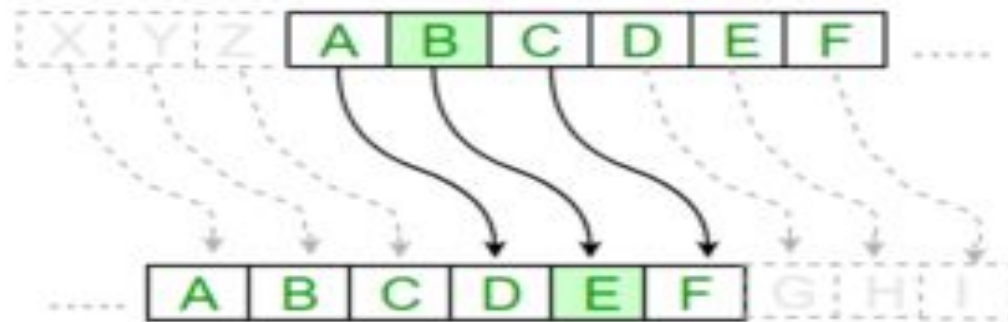
Conventional Cryptographic Techniques

Substitution Cipher

- Ceasor Cipher
- Monoalphabetic Cipher
- Polyalphabetic Cipher

Cesar Cipher

- The encryption can be represented using modular arithmetic by first transforming the letters into numbers, according to the scheme, $A = 0, B = 1, \dots, Z = 25$.



- **Plain Text:** ABCDEFGHIJKLMNOPQRSTUVWXYZ
Key: 4
Output: EFGHIJKLMNOPQRSTUVWXYZabcd
- **Plain Text:** I am studying Data Encryption
Key: 4
Output: M eq wxyhCmrk Hexe IrgvCtxmsr

- **Algorithm for Substitution Cipher:**
- **Input:**
- A String of both lower and upper case letters, called PlainText.
- An Integer denoting the required key.
- **Procedure:**
- Create a list of all the characters.
- Create a dictionary to store the substitution for all characters.
- For each character, transform the given character as per the rule, depending on whether we're encrypting or decrypting the text.
- Print the new string generated.

Monoalphabetic Cipher

A monoalphabetic cipher is any cipher in which the letters of the plain text are mapped to cipher text letters based on a single alphabetic key.

Plaintext Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M
Ciphertext Alphabet	Q	W	E	R	T	Y	U	I	O	P	A	S	D
Plaintext Alphabet	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext Alphabet	F	G	H	J	K	L	Z	X	C	V	B	N	M

Polyalphabetic Cipher

- A polyalphabetic cipher is any cipher based on substitution, using multiple substitution alphabets.
- Example:
- Consider the Plain Text: ATTACKATDAWN
- Keyword: LEMON

- Step 1: Convert the Alphabets into number.

Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M
Number	0	1	2	3	4	5	6	7	8	9	10	11	12
Alphabet	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Number	13	14	15	16	17	18	19	20	21	22	23	24	25

- Plaintext:

A	T	T	A	C	K	A	T	D	A	W	N
0	19	19	0	2	10	0	19	3	0	22	13

- Keyword (repeating):

L	E	M	O	N	L	E	M	O	N	L	E
11	4	12	14	13	11	4	12	14	13	11	4

- **Step 2: Encrypt Using:** $C_i = (P_i + K_i) \bmod 26$

A	T	T	A	C	K	A	T	D	A	W	N	L	E	M	O	N	L	E	M	O	N	L	E
0	19	19	0	2	10	0	19	3	0	22	13	11	4	12	14	13	11	4	12	14	13	11	4

P (Plaintext)	K (Keyword)	Formula	Result (C)	Cipher Letter
0 (A)	11 (L)	$(\underline{0} + 11) \bmod 26$	11 ✓	L ✓

P (Plaintext)	K (Keyword)	Formula	Result (C)	Cipher Letter
0 (A)	11 (L)	$(0 + 11) \bmod 26$	11	L
19 (T)	4 (E)	$(19 + 4) \bmod 26$	23	X
19 (T)	12 (M)	$(19 + 12) \bmod 26$	5	F
0 (A)	14 (O)	$(0 + 14) \bmod 26$	14	O
2 (C)	13 (N)	$(2 + 13) \bmod 26$	15	P
10 (K)	11 (L)	$(10 + 11) \bmod 26$	21	V

- **Cipher Text: LXFOPEFRNHR**
- **Step 3: Decrypt Using:** $P_i = (C_i - K_i + 26) \bmod 26$ and keyword **LEMON**

L	X	F	O	P	V	E	F	R	N	H	R	E	M	O	N	L	E	M	O	N	L	E
11	23	5	14	15	21	4	5	17	13	7	17	4	12	14	13	11	4	12	14	13	11	4

Cipher Letter (C)	Value (C)	Keyword (K)	Value (K)	$(C - K + 26) \bmod 26$	Result (P)	Plaintext Letter
E	4	E	4	$(4 - 4 + 26) \bmod 26 = 0$	0	A
F	5	M	12	$(5 - 12 + 26) \bmod 26 = 19$	19	T
R	17	O	14	$(17 - 14 + 26) \bmod 26 = 3$	3	D
N	13	N	13	$(13 - 13 + 26) \bmod 26 = 0$	0	A
H	7	L	11	$(7 - 11 + 26) \bmod 26 = 22$	22	W
R	17	E	4	$(17 - 4 + 26) \bmod 26 = 13$	13	N

- **Plaintext: ATTACKATDAWN**
- **Keyword: LEMON**
- **Cipher Text: LXFOPVEFRNHR**
- **Decrypted Text: ATTACKATDAWN**

Monoalphabetic Cipher	Polyalphabetic Cipher
A monoalphabetic cipher is one where each symbol in plain text is mapped to a fixed symbol in cipher text.	Polyalphabetic cipher is any cipher based on substitution, using multiple substitution alphabets.
The relationship between a character in the plain text and the characters in the cipher text is one-to-one.	The relationship between a character in the plain text and the characters in the cipher text is one-to-many.
Each alphabetic character of plain text is mapped onto a unique alphabetic character of a cipher text.	Each alphabetic character of plain text can be mapped onto 'm' alphabetic characters of a cipher text.
A stream cipher is a monoalphabetic cipher if the value of key does not depend on the position of the plain text character in the plain text stream.	A stream cipher is a polyalphabetic cipher if the value of key does depend on the position of the plain text character in the plain text stream.
It includes additive, multiplicative, affine and monoalphabetic substitution cipher.	It includes autokey , Playfair, Vigenere, Hill, one-time pad, rotor, and Enigma cipher.
It is a simple substitution cipher.	It is multiple substitutions cipher.
Monoalphabetic Cipher is described as a substitution cipher in which the same fixed mappings from plain text to cipher letters across the entire text are used.	Polyalphabetic Cipher is described as substitution cipher in which plain text letters in different positions are enciphered using different cryptoalphabets.
Monoalphabetic ciphers are not that strong as compared to polyalphabetic cipher.	Polyalphabetic ciphers are much stronger

Playfair cipher

- The **Playfair cipher** was the first practical digraph substitution cipher.
- The scheme was invented in **1854** by **Charles Wheatstone** but was named after Lord Playfair who promoted the use of the cipher.
- In playfair cipher unlike [traditional cipher](#) we encrypt a pair of alphabets(digraphs) instead of a single alphabet.
- It was used for tactical purposes by British forces in the Second Boer War and in World War I and for the same purpose by the Australians during World War II.
- This was because Playfair is reasonably fast to use and requires no special equipment.

- *The algorithm consists of 2 steps:*
- ***Generate the key Square (5x5):***
 - *The key square is a 5×5 grid of alphabets that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be unique and one letter of the alphabet (usually J) is omitted from the table (as the table can hold only 25 alphabets). If the plaintext contains J, then it is replaced by I.*
 - *The initial alphabets in the key square are the unique alphabets of the key in the order in which they appear followed by the remaining letters of the alphabet in order.*
- ***Algorithm to encrypt the plain text:*** *The plaintext is split into pairs of two letters (digraphs). If there is an odd number of letters, a Z is added to the last letter.*

Examples

- **PlainText:** "instruments"
After Split: 'in' 'st' 'ru' 'me' 'nt' 'sz'
Explanation: Pair cannot be made with same letter. Break the letter in single and add a bogus letter to the previous letter. Here 'z' is the bogus letter.
- **Plain Text:** "hello"
After Split: 'he' 'lx' 'lo'
Explanation: Here 'x' is the bogus letter.
- **Plain Text:** "helloe"
After Split: 'he' 'lx' 'lo' 'ez'
Explanation: If the letter is standing alone in the process of pairing, then add an extra bogus letter with the alone letter. Here 'x' and 'z' are the bogus letters.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Rules for Encryption

- ***If both the letters are in the same column:***
Take the letter below each one (going back to the top if at the bottom).

- ***For example:***

Diagraph: "me"

Encrypted Text: cl

Encryption: m -> c e -> l

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- ***If both the letters are in the same row:*** Take the letter to the right of each one (going back to the leftmost if at the rightmost position).

- ***For example:***

Diagraph: "st"

Encrypted Text: tl

Encryption: s -> t t -> l

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- ***If neither of the above rules is true: Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.***

- ***For example:***

Diagraph: "nt"

Encrypted Text: rq

Encryption: n -> r t -> q

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- **For example:**
Plain Text: "instrumentsz"
Encrypted Text: gatlmzclrqtx
Encryption:

i -> *g*

n -> *a*

s -> *t*

t -> *l*

r -> *m*

u -> *z*

m -> *c*

e -> *l*

n -> *r*

t -> *q*

s -> *t*

z -> *x*

•

in:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

st:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

ru:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

me:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

nt:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

sz:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Rules for Decryption

- If both the letters are in the same column: Take the letter above each one (going back to the bottom if at the top).*

For example:

Diagraph: "cl"

Decrypted Text: me

Decryption: c -> m l -> e

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- ***If both the letters are in the same row:*** Take the letter to the left of each one (going back to the rightmost if at the leftmost position).

For example:

Diagraph: "t/"

Decrypted Text: st

Decryption: t -> s / -> t

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- ***If neither of the above rules is true:*** Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.

For example:

Diagraph: "rq"

Decrypted Text: nt

Decryption: $r \rightarrow n$ $q \rightarrow t$

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- **For example:**

Plain Text: "gatlmzclrqtz"

Decrypted Text: instrumentsz

Decryption:

(red) -> (green)

ga -> in

tl -> st

mz -> ru

cl -> me

rq -> nt

tx -> sz

in:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

st:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

ru:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

me:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

nt:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

sz:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Hill Cipher

- Hill cipher is a polygraphic substitution cipher based on linear algebra. Each letter is represented by a number modulo 26.
- Often the simple scheme $A = 0, B = 1, \dots, Z = 25$ is used, but this is not an essential feature of the cipher.
- To encrypt a message, each block of n letters (considered as an n -component vector) is multiplied by an invertible $n \times n$ matrix, against modulus 26.
- To decrypt the message, each block is multiplied by the inverse of the matrix used for encryption.
- The matrix used for encryption is the cipher key, and it should be chosen randomly from the set of invertible $n \times n$ matrices (modulo 26).

Example

- Input : Plaintext: ACT
- Key: GYBNQKURP
- Output : Ciphertext: POH

- **Encryption**

- We have to encrypt the message 'ACT' (n=3). The key is 'GYBNQKURP' which can be written as the nxn

matrix:

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

The message 'ACT' is written as vector:

$$\begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix}$$

- The enciphered vector is given as:

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} = \begin{bmatrix} 67 \\ 222 \\ 319 \end{bmatrix} \equiv \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix} \pmod{26}$$

which corresponds to ciphertext of 'POH'

- **Decryption**

- To decrypt the message, we turn the ciphertext back into a vector, then simply multiply by the inverse matrix of the key matrix (IFKVIVVMI in letters).
- The inverse of the matrix used in the previous example is:

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}^{-1} \equiv \begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix} \pmod{26}$$

- For the previous Ciphertext 'POH':

$$\begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix} \equiv \begin{bmatrix} 260 \\ 574 \\ 539 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} \pmod{26}$$

which gives us back 'ACT'.

Assume that all the alphabets are in upper case.

Below is the implementation of the above idea for n=3.

One Time Pad

The two requirements for the One-Time pad are:

- The key should be **randomly generated as long as the size of the message.**
- The key is to be used to encrypt and decrypt a **single message, and then it is discarded.**

Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M
Number	0	1	2	3	4	5	6	7	8	9	10	11	12
Alphabet	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Number	13	14	15	16	17	18	19	20	21	22	23	24	25

Input: Message = HELLO,

Key = MONEY

Output: Cipher - TSYPM,

Message - HELLO

Explanation: Part 1: Plain text to Ciphertext

Plain text — H E L L O ? 7 4 11 11 14

Key — M O N E Y ? 12 14 13 4 24

Plain text + key ? 19 18 24 15 38

Cipher Text ? T S Y P M

Part 2: Ciphertext to Message

Cipher Text — T S Y P M ? 19 18 24 15 12

Key — M O N E Y ? 12 14 13 4 24

Cipher text - key ? 7 4 11 11 -12 ? 7 4 11 11 14

Message ? H E L L O

Transposition Cipher

- The Transposition Cipher Technique is an encryption method used to encrypt a message or information.
- This encryption method is done by playing with the position of letters of the plain text.
- The positions of the characters present in the plaintext are rearranged or shifted to form the ciphertext.
- It makes use of some kind of permutation function to achieve the encryption purpose.
- It is very easy to use and so simple to implement.

Types of Transposition Cipher Techniques

- Rail Fence Transposition Cipher
- Block (Single Columnar) Transposition Cipher
- Double Columnar Transposition Cipher

Rail Fence Transposition Cipher

- Rail Fence Transposition cipher technique is the simplest transposition cipher technique.
- It is also termed as a zigzag cipher.
- It gets its name from the way through which it performs encryption of plain text.

Plain text - Good morning
depth - 2

① G o m r i g
② o d o n n x

Cipher text $\rightarrow$ Gomrigo donnx (length = 12)
depth $\rightarrow$ 2

PT : $\frac{12}{2} = 6$

Gomrig	odonnx
--------	--------

Diagram showing the cipher text "Gomrig" and "odonnx" separated by a vertical line. Two diagonal lines cross the vertical line, and two vertical lines extend downwards from the "Gomrig" section, indicating the extraction of the first six characters of the plain text.

Plain text : Good morning/x

Block (Single Columnar) Transposition Cipher

- Plaintext: HELLOWORLD

- Key: CAT

- Step 1: Number the key letters by alphabetical order

Letter	C	A	T
Order	2	1	3

- **Step 2: Write plaintext in rows under each key letter**
- The plaintext has 10 letters and the key length is 3, we'll write it in rows of 3 columns (pad if needed).

HELLOWORLD

C	A	T
H	E	L
L	O	W
O	R	L
D	X	X

- **Step 3: Read columns in key order: 1, 2, 3**

Letter	C	A	T
Order	2	1	3

C	A	T
H	E	L
L	O	W
O	R	L
D	X	X

Order	Column letters
1 (A)	E O R X
2 (C)	H L O D
3 (T)	L W L X

Step 4: Write ciphertext by concatenating columns in order

Ciphertext: EORXHLODLWLX

Decryption

- Ciphertext: EORXHL~~ODL~~WLX
- Key: CAT
- Step 1: Number the key letters by alphabetical order

Letter	C	A	T
Order	2	1	3

- Ciphertext: EORX~~H~~LODLWLX
- Key: CAT
- **Step 2:** divide the ciphertext column-wise based on key order
- Split the ciphertext into 3 columns, each with 4 letters:
- A (1st in order): E O R X
- C (2nd in order): H L O D ✓
- T (3rd in order): L W L X

- **Step 3: Read row-wise for decryption**

Letter	C	A	T
Order	2	1	3

C	A	T	Decrypted Text
H	E	L	
L	O	W	
O	R	L	
D	X	X	

- **Step 3: Read row-wise for decryption**

Letter	C	A	T
Order	2	1	3

C	A	T	Decrypted Text
H	E	L	HEL
L	O	W	LOW
O	R	L	ORL
D	X	X	DXX

Final Decrypted Plaintext:

HELLOWORLD

(Padded X's can be discarded)

- Fill the grid row by row using Key 1. Removing spaces and punctuation makes a tighter grid, but the tool will also accept them if you keep them.
- Key 1: FIRM Plaintext: MEETMEATDAWN

F	I	R	M
1	2	4	3
M	E	E	T
M	E	A	T
D	A	W	N

- **Read columns in Key 1 order**
- Sort the key alphabetically (ties keep their left-to-right order) and read the columns in that order to form the intermediate text.
- Key 1 order: F, I, M, R
- Intermediate text: **MMDEEATTNEAW**

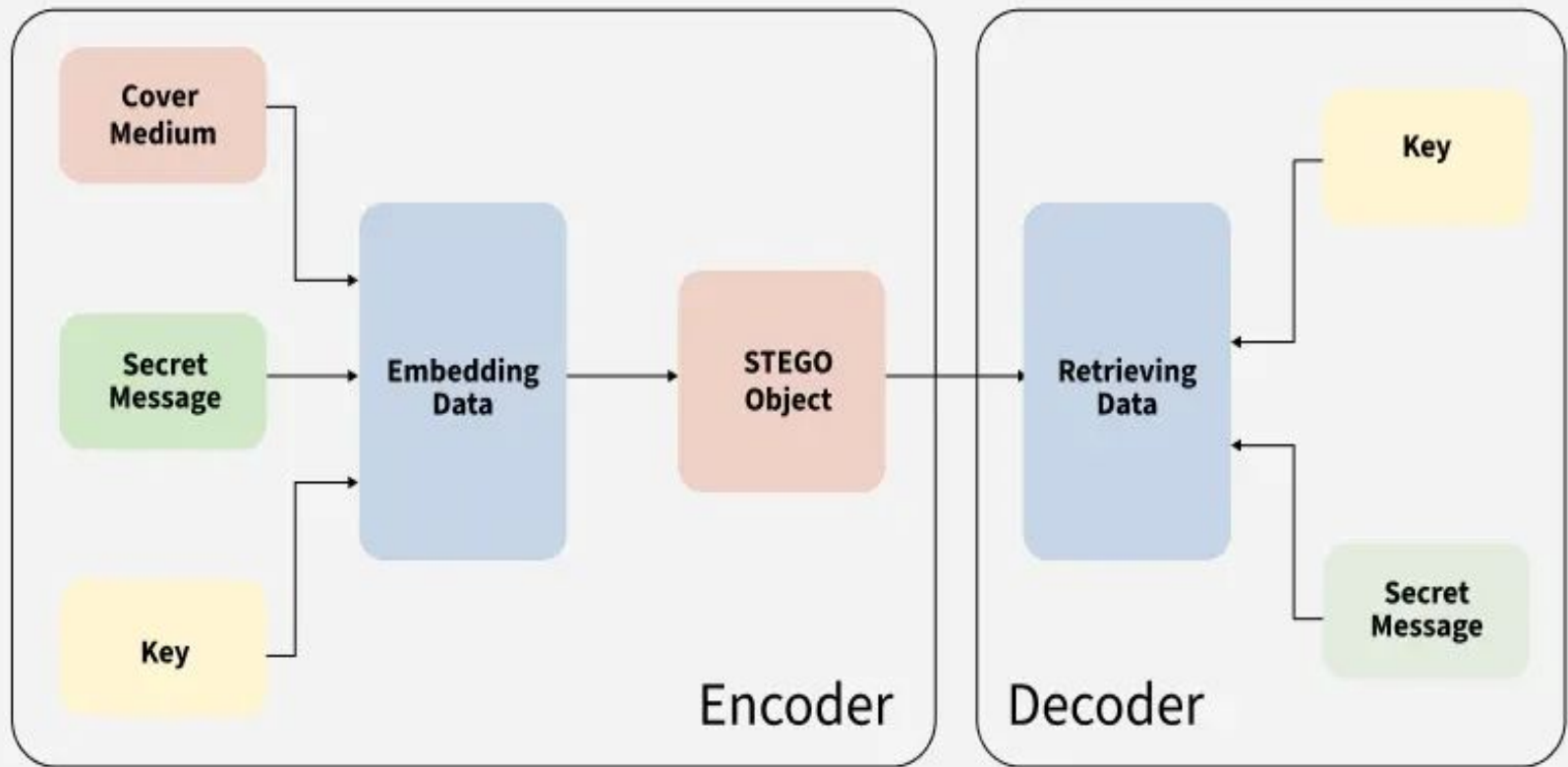
- **Write the intermediate text under Key 2**
- Repeat the same grid process using Key 2 and the intermediate text from the first pass.
- Key 2: WORD
- Intermediate text: MMDEEATTNEAW

W	O	R	D
4	2	3	1
M	M	D	E
E	A	T	T
N	E	A	W

- **Read columns in Key 2 order**
- Read the Key 2 columns in order to produce the final ciphertext.
- Key 2 order: D, O, R, W Ciphertext: ETWMAEDTAMEN
- **Decrypt by reversing the order**
- To decrypt, rebuild the Key 2 grid column-by-column to recover the intermediate text, then repeat with Key 1 to recover the plaintext.
- Ciphertext: ETWMAEDTAMEN
- After Key 2: MMDEEATTNEAW
- After Key 1: MEETMEATDAWN

steganography

- The term "steganography" is derived from the Greek word "steganos" which means "hidden or covered" and "graph" means "to write."
- Steganography is the practice of concealing information. It involves hiding data within an ordinary, non-secret file or message to prevent detection.
- The hidden information is being extracted at the receiving end.



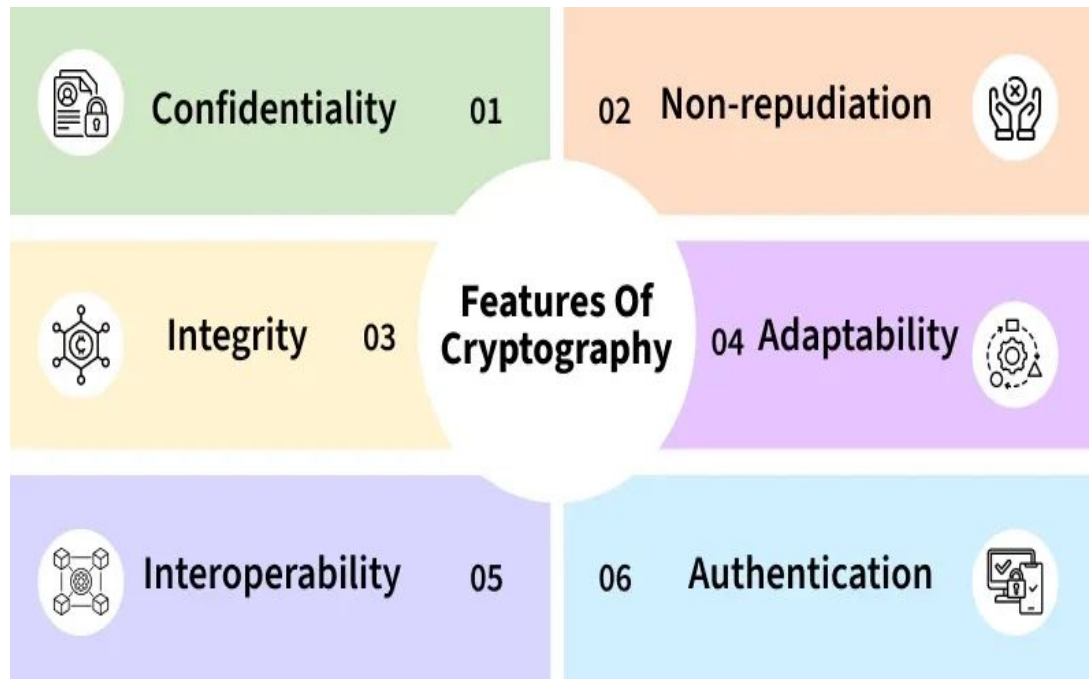
- **Step 1:** The first step in steganography is selecting a cover medium which is the file or message that will carry the hidden data. Common cover media include:
 - Images (JPEG, PNG, BMP, etc.)
 - Audio files (MP3, WAV, etc.)
 - Video files (MP4, AVI, etc.)
 - Text files or documents
- **Step 2:** Sometimes, before embedding, the secret message is encrypted to add an additional layer of security. This ensures that even if someone detects the hidden data, they cannot read it without the decryption key.

- **Step 3:** The secret message is then hidden using one of several techniques:
- **Least Significant Bit (LSB):** The least significant bit of a byte is changed to hide the secret message. This method is often used in image and audio files.
- **Frequency Domain:** Instead of modifying the raw data (like pixels or audio samples), the secret message can be embedded in the frequency components of an image or audio file.
- **Bit Planes:** In this method, data is hidden in the higher-order bit planes of an image. This can be more secure because it uses bits that are less likely to be noticed.
- The most common is **Least Significant Bit (LSB)** encoding.

Step 4: The modified data is then embedded into the cover medium. The resulting file which now contains both the cover data and the hidden message is referred to as the **stego-object** which can be safely transmitted or stored without raising suspicion.

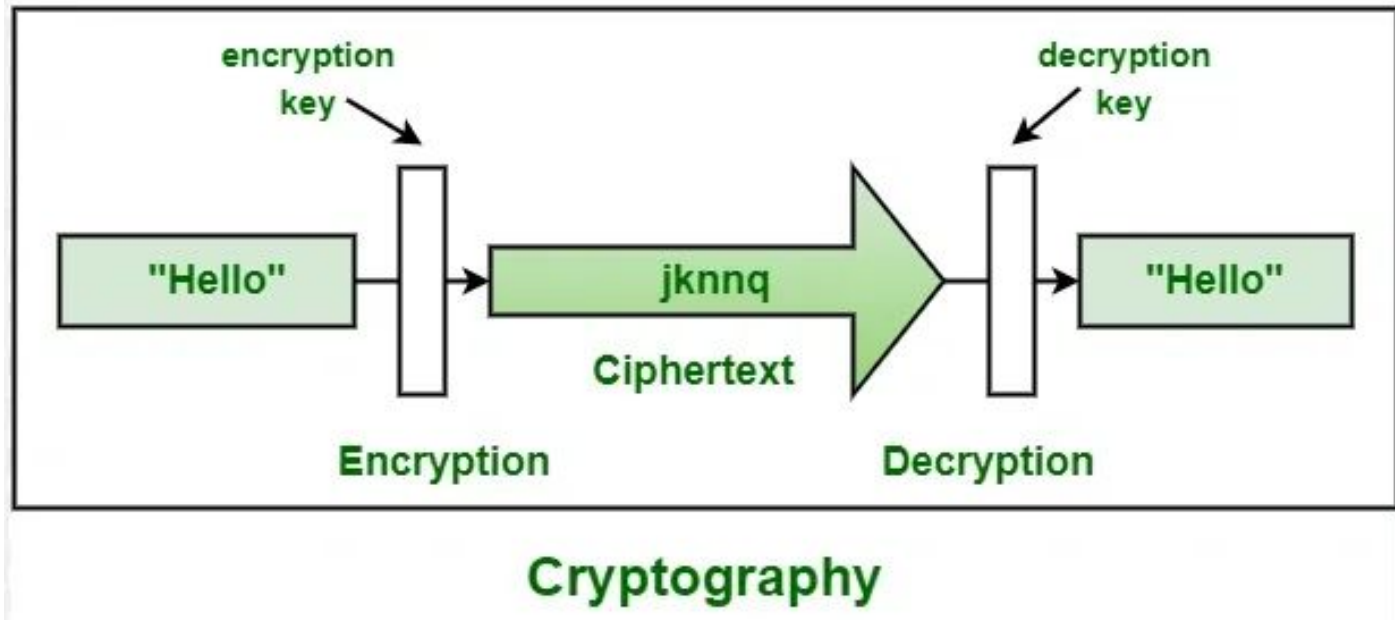
Step 5: The receiver of the stego-object needs to know the method used for embedding the secret message. In some cases, a secret key is required to extract the data if encryption is used in combination with steganography.

Cryptography

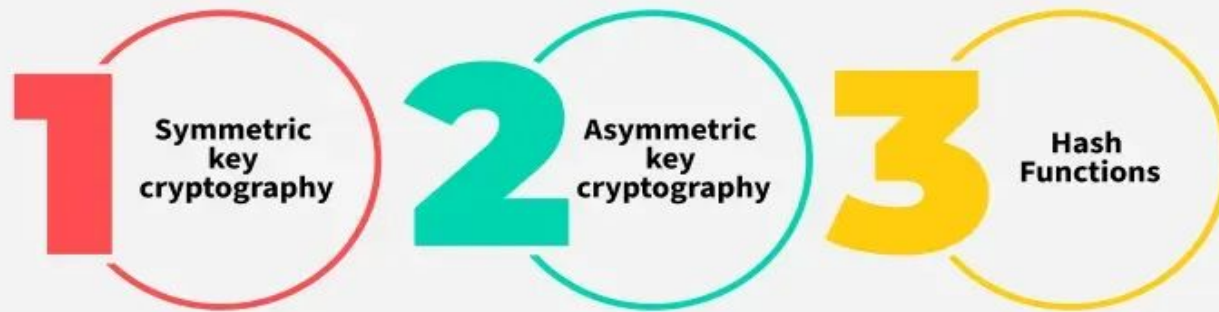


- **Confidentiality:** Information can only be accessed by the person for whom it is intended and no other person except him can access it.
- **Non-repudiation:** The creator/sender of information cannot deny his intention to send information at a later stage.
- **Integrity:** Information cannot be modified in storage or transition between sender and intended receiver without any addition to information being detected.
- **Adaptability:** Cryptography continuously evolves to stay ahead of security threats and technological advancements.
- **Interoperability:** Cryptography allows for secure communication between different systems and platforms.
- **Authentication:** The identities of the sender and receiver are confirmed. As well destination/origin of the information is confirmed.

Working of a cryptography



Types of Cryptography



Applications of cryptography

Computer
Passwords



Digital
Currencies



Secure Web
Browsing



Electronic
Signatures



Authentication



Cryptocurrencies



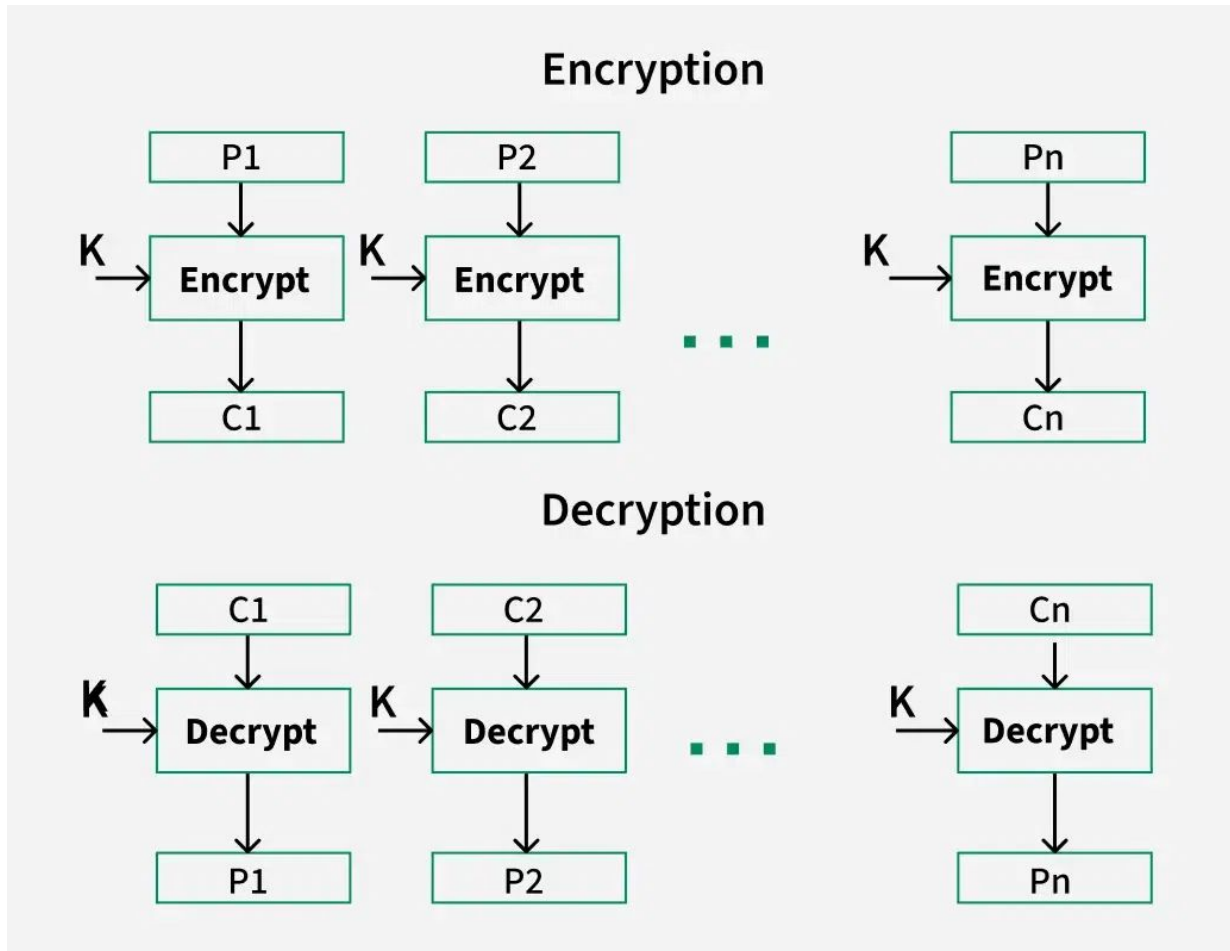
End-to-end Internet
Encryption



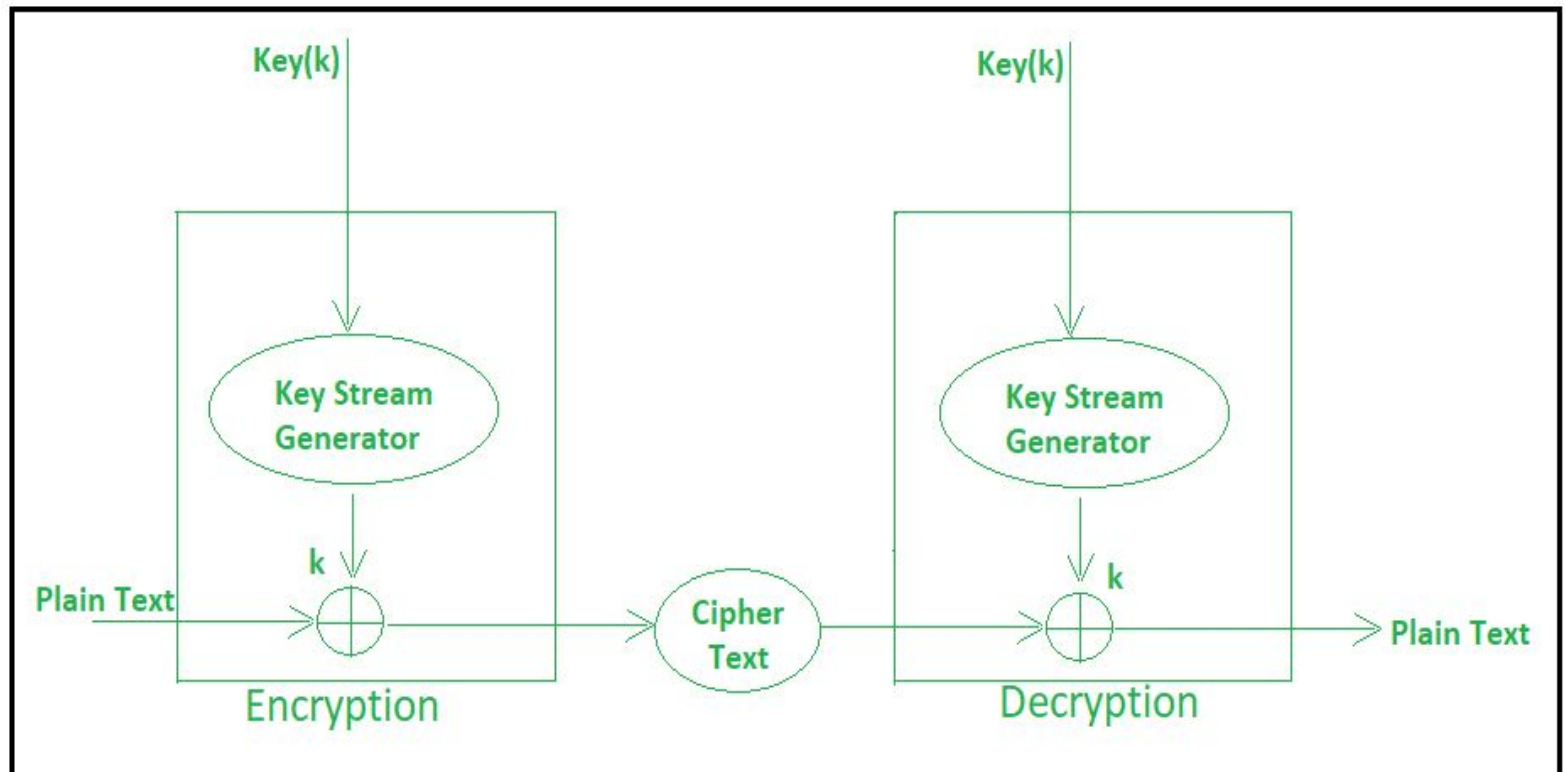
Block and Stream Cipher

- **Block ciphers** encrypt data in fixed-size chunks (blocks), like 128 bits, using complex substitutions and permutations (e.g., AES), good for large data storage;
- **Stream ciphers** encrypt bit-by-bit or byte-by-byte, generating a keystream XORed with plaintext (e.g., [RC4](#)), ideal for real-time, continuous data streams

Block Cipher



Stream Cipher

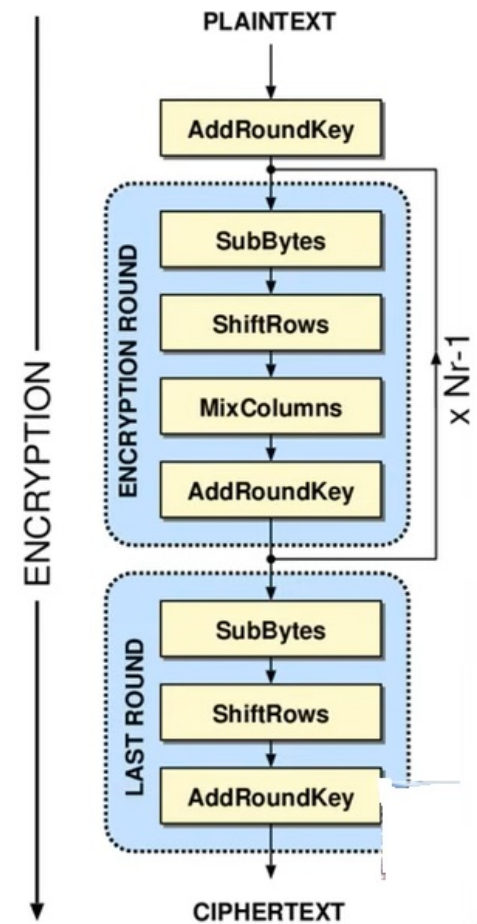


Data Encryption Standard

- Data Encryption Standard (DES) is a symmetric block cipher. By 'symmetric', we mean that the size of input text and output text (ciphertext) is same (64-bits).
- The 'block' here means that it takes group of bits together as input instead of encrypting the text bit by bit.
- Data encryption standard (DES) has been found vulnerable to very powerful attacks and therefore, it was replaced by Advanced Encryption Standard (AES).

AES (Advanced Encryption Standard)

- AES uses 128 bits of plain text.
- AES consists of multiple rounds of processing different key bits like,
 - 128 bits key – perform 10 rounds of encryption
 - 192 bits key – perform 12 rounds of encryption.
 - 256 bits key – perform 14 rounds of encryption.
- Overall structure of AES encryption process shown in figure.
- For encryption, each round consists of the following four steps:
 - 1) Sub Bytes – 10 Rounds
 - 2) Shift Rows – 10 Rounds
 - 3) Mix Columns – 9 Rounds (Not present in last round)
 - 4) Add Round Key – 10 Rounds
- Above steps also called *AES transformation function*.



AES Subbytes

- First transformation function of AES is called as Substitute Bytes (Sub Bytes).
- It is also known as forward substitute bytes transformation.
- AES defines a 16 x 16 matrix of byte values, called an **S-box**, that contains a permutation of all possible 256 8-bit values.
- These row and column values serve as indexes into the S-box to select a unique 8-bit output value.

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	CB	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

AES Sub Bytes



	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

1st Number E → Row
2nd Number A → Column

EA

EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5



?			

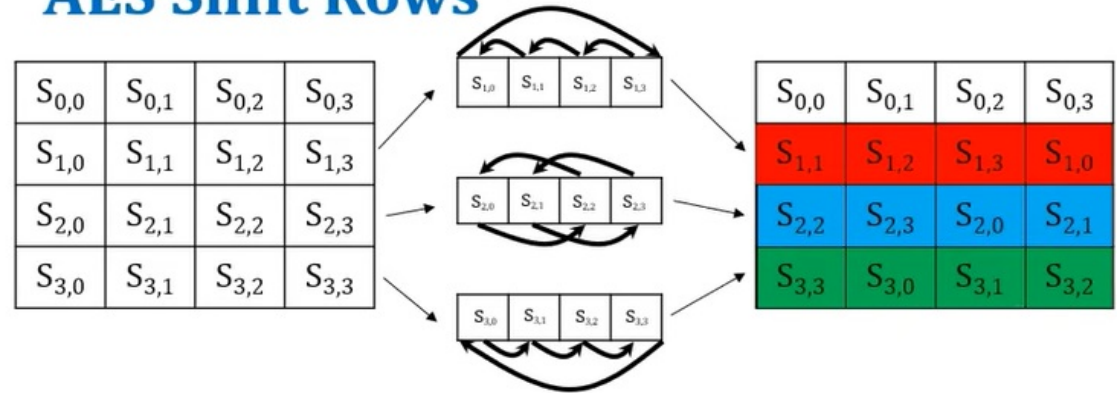
EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5



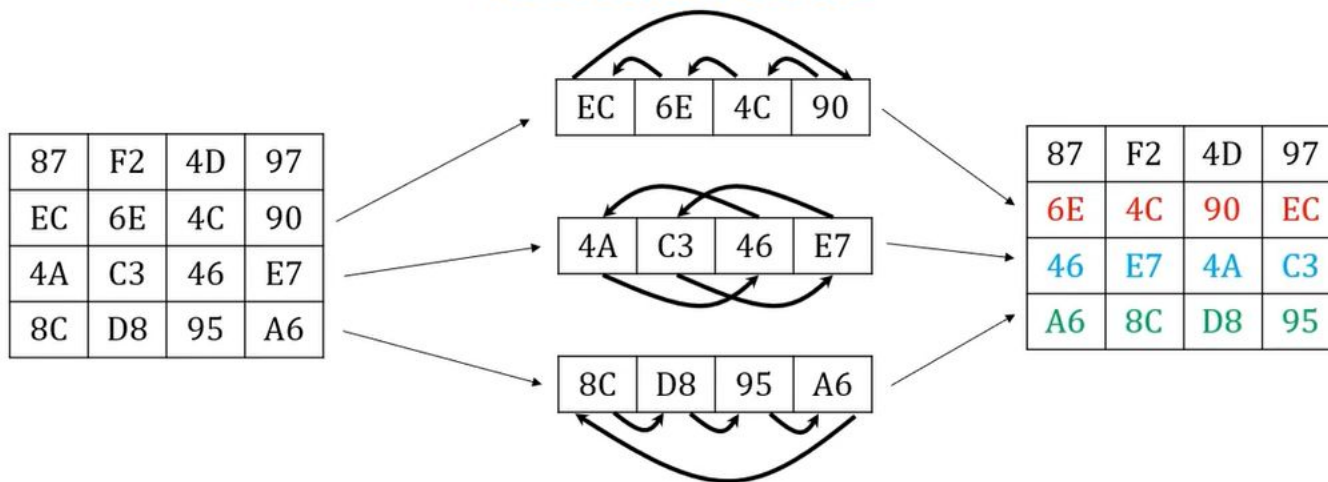
87	F2	4D	97
EC	6E	4C	90
4A	C3	46	E7
8C	D8	95	A6

- Second transformation function of AES is called as Shift Rows.
- It is also called as forward shift row transformation.
- State Array = Output of Substitute Bytes
- Rules of shifting rows,
 - Row 1 → No Shifting
 - Row 2 → 1 byte circular left shift
 - Row 3 → 2 byte circular left shift
 - Row 4 → 3 byte circular left shift

AES Shift Rows



AES Shift Rows



AES Mix Column

- Third transformation function of AES is called as Mix Columns, operates on each column individually.

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

*

$S_{0,0}$	$S_{0,1}$	$S_{0,2}$	$S_{0,3}$
$S_{1,0}$	$S_{1,1}$	$S_{1,2}$	$S_{1,3}$
$S_{2,0}$	$S_{2,1}$	$S_{2,2}$	$S_{2,3}$
$S_{3,0}$	$S_{3,1}$	$S_{3,2}$	$S_{3,3}$

=

$S'_{0,0}$	$S'_{0,1}$	$S'_{0,2}$	$S'_{0,3}$
$S'_{1,0}$	$S'_{1,1}$	$S'_{1,2}$	$S'_{1,3}$
$S'_{2,0}$	$S'_{2,1}$	$S'_{2,2}$	$S'_{2,3}$
$S'_{3,0}$	$S'_{3,1}$	$S'_{3,2}$	$S'_{3,3}$

Predefine Matrix

State Array

New State Array

- State Array = Output of Shift Rows function.

- Each byte of a column is mapped into a new value that is a function of all four bytes in that column.

$$S'_{0,j} = (2 * S_{0,j}) \oplus (3 * S_{1,j}) \oplus S_{2,j} \oplus S_{3,j}$$

$$S'_{1,j} = S_{0,j} \oplus (2 * S_{1,j}) \oplus (3 * S_{2,j}) \oplus S_{3,j}$$

$$S'_{2,j} = S_{0,j} \oplus S_{1,j} \oplus (2 * S_{2,j}) \oplus (3 * S_{3,j})$$

$$S'_{3,j} = (3 * S_{0,j}) \oplus S_{1,j} \oplus S_{2,j} \oplus (2 * S_{3,j})$$

AES Mix Column

<u>2</u>	3	1	1
1	2	3	1
1	1	2	3
3	1	1	2

*

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

=

?			

$$\{02\} * \{87\} \oplus \{03\} * \{6E\} \oplus \{01\} * \{46\} \oplus \{01\} * \{A6\}$$

$$X^7 + X^6 + X^5 + X^4 + X^3 + X^2 + X^1 + 1$$

$$\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{matrix}$$

$02 = 0000\ 0010 = X$
 $87 = 1000\ 0111 = X^7 + X^2 + X + 1$

$$\begin{matrix} X^7 & + & X^6 & + & X^5 & + & X^4 & + & X^3 & + & X^2 & + & X^1 & + & 1 \\ 1 & & 0 & & 0 & & 0 & & 0 & & 1 & & 1 & & 1 \end{matrix}$$

$$\begin{aligned}
 \{02\} * \{87\} &= X * (X^7 + X^2 + X + 1) \\
 &= X^8 + X^3 + X^2 + X \\
 &= X^4 + \cancel{X^3} + \cancel{X} + 1 + \cancel{X^3} + X^2 + \cancel{X} \\
 &= X^4 + X^2 + 1 \\
 &= 0001\ 0101
 \end{aligned}$$

Use irreducible Polynomial Theorem, GF (2⁸)
 $X^8 = X^4 + X^3 + X + 1$

$$\begin{matrix} X^7 & + & X^6 & + & X^5 & + & X^4 & + & X^3 & + & X^2 & + & X^1 & + & 1 \\ 0 & & 0 & & 0 & & 1 & & 0 & & 1 & & 0 & & 1 \end{matrix}$$

AES Mix Column

2	3	1	1
1	2	3	1
1	1	2	3
3	1	1	2

*

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

=

?			

$$\{02\} * \{87\} \oplus \{03\} * \{6E\} \oplus \{01\} * \{46\} \oplus \{01\} * \{A6\}$$

$$02 = 0000\ 0010 = X$$

$$87 = 1000\ 0111 = X^7 + X^2 + X + 1$$

$$\begin{aligned} 02 * 87 &= X * (X^7 + X^2 + X + 1) \\ &= X^8 + X^3 + X^2 + X \\ &= X^4 + \cancel{X^3} + \cancel{X} + 1 + \cancel{X^3} + X^2 + \cancel{X} \\ &= X^4 + X^2 + 1 \\ &= 0001\ 0101 \end{aligned}$$

$$03 = 0000\ 0011 = X + 1$$

$$6E = 0110\ 1110 = X^6 + X^5 + X^3 + X^2 + X$$

$$\begin{aligned} 03 * 6E &= (X+1) * (X^6 + X^5 + X^3 + X^2 + X) \\ &= X^7 + \cancel{X^6} + X^4 + \cancel{X^3} + \cancel{X^2} + \cancel{X^6} + X^5 + \cancel{X^3} + \cancel{X^2} + X \\ &= X^7 + X^5 + X^4 + X \\ &= 1011\ 0010 \end{aligned}$$

$$01 * 46 = 46 = 0100\ 0110$$

$$01 * A6 = A6 = 1010\ 0110$$

AES Mix Column

2	3	1	1
1	2	3	1
1	1	2	3
3	1	1	2

*

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

=

?			

$$\{02\} * \{87\} \oplus \{03\} * \{6E\} \oplus \{01\} * \{46\} \oplus \{01\} * \{A6\}$$

$$02 * 87 = 00010101$$

$$03 * 6E = 10110010$$

$$01 * 46 = 01000110$$

$$01 * A6 = 10100110$$

$$= \begin{array}{cccc} 0 & 1 & 0 & 0 \\ \hline & 4 & & \end{array} \begin{array}{cccc} 0 & 1 & 1 & 1 \\ \hline & 7 & & \end{array} = \{47\}$$



In Ex-or

Odd time 1's = 1

Even time 1's = 0

AES Mix Column

2	3	1	1
<u>1</u>	2	3	1
1	1	2	3
3	1	1	2

*

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

=

47			
37			

$$\{01\} * \{87\} \oplus \{02\} * \{6E\} \oplus \{03\} * \{46\} \oplus \{01\} * \{A6\} = \{37\}$$

$$01 * 87 = 10000111$$

$$02 * 6E = 11011100$$

$$03 * 46 = 11001010$$

$$01 * A6 = 10100110$$

$$\begin{array}{r}
 10000111 \\
 11011100 \\
 11001010 \\
 10100110 \\
 \hline
 0011 \quad 0111 \\
 \hline
 3 \quad 7
 \end{array}
 = \{37\}$$

AES Mix Column

2	3	1	1
1	2	3	1
1	1	2	3
3	1	1	2

*

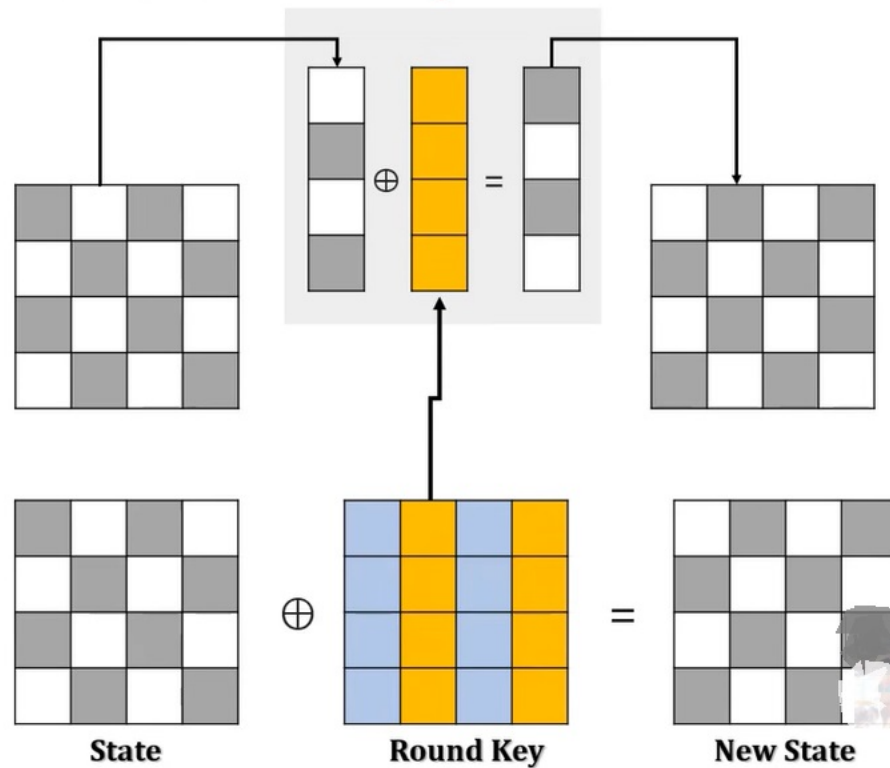
87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

=

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

AES Add Round key

- Fourth transformation function of AES is called as Add Round Key.
- It is also known as forward add round key transformation.
- The 128 bits of State are bitwise XORed with the 128 bits of the round key.
- Add Round Key proceeds one column at a time.



AES Add Round key

47	40	A3	4C	$\oplus$	AC	19	28	57	=	EB			
37	D4	70	9F		77	FA	D1	5C		40			
94	E4	3A	42		66	DC	29	00		F2			
ED	A5	A6	BC		F3	21	41	6A		?			

$$47 \oplus AC = ?$$

$$47 = 0100\ 0111$$

$$AC = 1010\ 1100$$

$$\begin{array}{r} 0100\ 0111 \\ 1010\ 1100 \\ \hline 1110\ 1011 \end{array}$$

$$\begin{array}{cc} \text{E} & \text{B} \end{array} = \{\text{EB}\}$$

$$94 \oplus 66 = ?$$

$$94 = 1001\ 0100$$

$$66 = 0110\ 0110$$

$$\begin{array}{r} 1001\ 0100 \\ 0110\ 0110 \\ \hline 1111\ 0010 \end{array}$$

$$\begin{array}{cc} \text{F} & 2 \end{array} = \{\text{F2}\}$$

$$37 \oplus 77 = ?$$

$$37 = 0011\ 0111$$

$$77 = 0111\ 0111$$

$$\begin{array}{r} 0011\ 0111 \\ 0111\ 0111 \\ \hline 0100\ 0000 \end{array}$$

$$\begin{array}{cc} 4 & 0 \end{array} = \{40\}$$

AES Add Round key

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

$\oplus$

AC	19	28	57
77	FA	D1	5C
66	DC	29	00
F3	21	41	6A

=

EB	59	8B	1B
40	2E	A1	C3
F2	38	13	42
1E	84	E7	D6

$$4C \oplus 57 = ?$$

$$4C = 0100\ 1100$$

$$57 = 0101\ 0111$$

$$\begin{array}{r} 0100\ 1100 \\ 0101\ 0111 \\ \hline 0001\ 1011 \end{array}$$

$$\begin{array}{cc} 1 & B = \{1B\} \end{array}$$

$$42 \oplus 00 = ?$$

$$42 = 0100\ 0010$$

$$00 = 0000\ 0000$$

$$\begin{array}{r} 0100\ 0010 \\ 0000\ 0000 \\ \hline 0100\ 0010 \end{array}$$

$$\begin{array}{cc} 4 & 2 = \{42\} \end{array}$$

$$9F \oplus 5C = ?$$

$$9F = 1001\ 1111$$

$$5C = 0101\ 1100$$

$$\begin{array}{r} 1001\ 1111 \\ 0101\ 1100 \\ \hline 1100\ 0011 \end{array}$$

$$\begin{array}{cc} C & 3 = \{C3\} \end{array}$$

$$BC \oplus 6A = ?$$

$$BC = 1011\ 1100$$

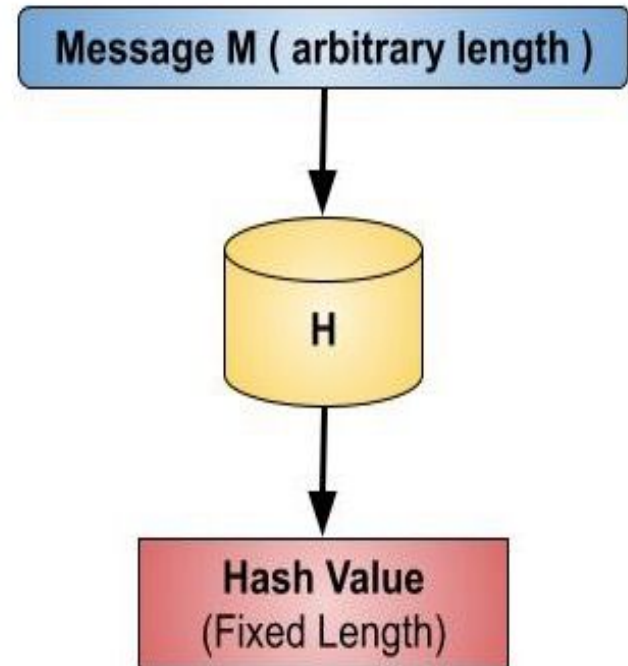
$$6A = 0110\ 1010$$

$$\begin{array}{r} 1011\ 1100 \\ 0110\ 1010 \\ \hline 1101\ 0110 \end{array}$$

$$\begin{array}{cc} D & 6 = \{D6\} \end{array}$$

Hash Function

- Hash functions are mathematical operations that "map" or change a given collection of data into a fixed-length bit string that is referred to as the "hash value."
- Hash functions have a variety of complexity and difficulty levels and are used in cryptography.
- Cryptocurrency, password security, and communication security all use hash functions.



- Cryptographic hash functions are mathematical algorithms that transform input data into a fixed-length sequence of characters, referred to as a hash value.
- Cryptographic hash functions are intended to be fast, deterministic, and one-way, meaning that even a minor change in input yields a very different hash.
- These functions are important for securing information digitally by allowing data verification and authentication.

Properties of Cryptographic Hash Functions

- **Deterministic:** The same input always generates the exact same hash output, ensuring consistent and reliable verification of data.
- **Fast Computation:** Cryptographic hash functions are designed to process inputs quickly and efficiently, making them practical for handling large datasets and real-time applications.
- **Pre-image Resistance:** It is computationally infeasible to reverse-engineer or retrieve the original input data from its hash value, protecting sensitive information from exposure.
- **Collision Resistance:** The function minimizes the chance that two distinct inputs will produce identical hash values, ensuring unique data fingerprints for security and integrity.
- **Avalanche Effect:** Even a tiny change in the input, such as flipping a single bit, causes a significant and unpredictable change in the hash output, enhancing the function's sensitivity to data modifications.

Popular Cryptographic Hash Algorithms

- **MD5 (Message Digest Algorithm 5)**

Once widely used for data integrity and digital signatures, MD5 is now considered insecure due to vulnerabilities that allow attackers to generate hash collisions easily. Its speed and simplicity made it popular historically, but it is no longer recommended for security-critical applications.

- **SHA-1 (Secure Hash Algorithm 1)**

SHA-1 improved upon MD5 with a longer hash length and better resistance to collisions. However, advances in computational power and cryptanalysis exposed weaknesses, leading to practical collision attacks. Consequently, SHA-1 is deprecated for most security uses, including SSL/TLS certificates and digital signatures.

- **SHA-2 Family (SHA-256, SHA-512)**

The SHA-2 family is currently the industry standard for cryptographic hashing, offering robust security with longer hash outputs of 256 and 512 bits. These algorithms provide strong collision and pre-image resistance, making them the preferred choice for secure communication protocols, blockchain technologies, and password hashing.

- **SHA-3 (Keccak)**

Adopted as the latest NIST standard, SHA-3 uses a unique sponge construction different from SHA-2, enhancing security and flexibility. It offers comparable hash lengths with improved resistance to certain types of attacks, making it suitable for applications demanding long-term security.

- **BLAKE2 & BLAKE3**

Designed as high-speed, secure alternatives to SHA-2 and SHA-3, BLAKE2 and BLAKE3 deliver faster hashing without compromising security. BLAKE3, in particular, supports parallel processing and incremental updates, making it ideal for modern systems requiring both speed and strong cryptographic guarantees.

Secure Hash Algorithm (SHA - 1)

□ *Introduction of Secure Hash Algorithm:*

- The secure hash algorithm (SHA) was developed by National Institute of Standards and Technology (NIST). It is based on MD4 algorithm. Based on different digest lengths, SHA includes algorithms such as SHA-1, SHA-256, SHA-384 and SHA-512.
- Unlike encryption, given a variable length message x , a secure hash algorithm computes a function $H(x)$ which has a fixed bits. When a message of any length is less than 2^{64} bits is input, the SHA-1 produces a 160-bit output called message digest.
- SHA-1 called secure because it is computationally infeasible to find a message which corresponds to a given message digest, or to find two different messages which produce the same message digest.
- The most commonly used hash function from the SHA family is SHA-1. SHA-1 is used in SSL/TLS, PGP, SSH, MIME and IPsec for security and authentication purpose.

Working of SHA – 1

- SHA1 works with any input message that is less than 2^{64} bits in length.
- The output of SHA is a message digest, which is 160 bits in length.

□ Working of SHA-1

➤ Step – 1: Padding

- The first step of SHA-1 is added padding to the end of original message to prepare message in multiple of 512 bits.

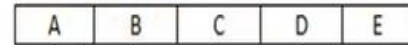
➤ Step – 2: Append Length:

- The length of message excluding the length of the padding is now calculated and appended to the end of the padding as 64-bit block. (message length is 64 bits short of multiple of 512).

Working of SHA – 1

➤ Step – 3: Divide the input into 512-bit blocks:

- The input message is now divided into blocks, each of length 512 bits.



➤ Step – 4: Initialize chaining variables:

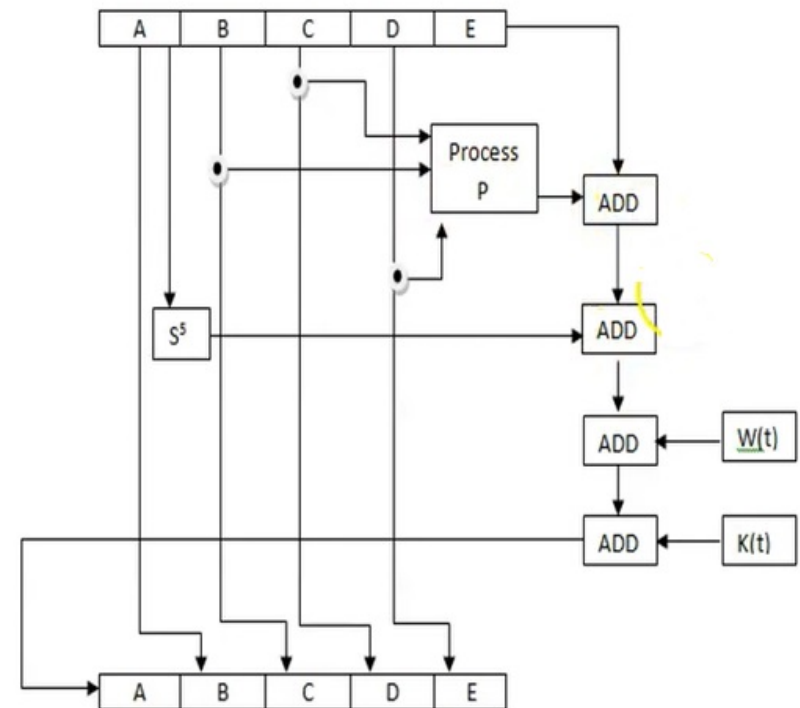
- Now, five chaining variables A to E are initialized.
- Each of 32 bits variable produces 160 bits length of message digest.

Working of SHA – 1

➤ Step – 5: Process Block & Output:

- Combination of A-E chaining variable is called ABCDE, will be considered as a single register.
- Now divided the current 512-bit block into 16 sub blocks, each consisting of 32 bits. (32x16=512)
- SHA-1 has perform four rounds.
- Each round takes the current 512-bit block, the register ABCDE and constant $K(t)$ (where $t=0$ to 79) as input.
- SHA consists of four rounds, each round containing 20 iteration. So total iteration is 80.
- The logical operation of a single SHA-1 iteration looks as shown in figure.
- Mathematically, an iteration consists of the following operation:

$$ABCDE = E + \text{Process } P + S^5(a) + W(t) + K(t)$$



W_t = expanded message word of round t ;

K_t = round constant of round t

Working of SHA-1

- **Initialization**
 - Let's start with some initial values.
 - Set all five variables (h0 to h4) with the correct values.
- **Pre-processing**
 - Prepare the message ready for hashing.
 - A '1' bit should be added to the message's end.
 - To set a maximum length for the message, append a few '0' bits.
 - At the end of the message, include the message's original length (in bits).
- **Processing**
 - Divide the message into manageable parts and deal with each one separately.
 - There should be 512 bits in each section.
 - Divide the pieces into sixteen 32-bit words each.
 - Use a formula to extend these words into eighty 32-bit words.

- **Main loop**

- Go through every word and do some calculations.
- Based on the position and values of the word, update the temporary variables.
- As per these calculations, update the hash values.

- **Finalization**

- To get the final hash, combine the hash values.
- Using a predetermined order, arrange the hash values to generate a 160-bit integer.

Comparison based on parameter of different SHA version

- SHA – 1 was cracked in the year 2005. New hash function SHA-512 is introduced to overcome problem SHA-1.

❖ Comparison of SHA Parameters:

Sr. No.	Parameters	SHA-1	SHA-256	SHA-384	SHA-512
1	Message digest size	160	256	384	512
2	Message size	$< 2^{64}$	$< 2^{64}$	$< 2^{128}$	$< 2^{128}$
3	Block size	512	512	1024	1024
4	Word size	32	32	64	64
5	Number of steps	80	64	80	80
6	Security level	80	128	192	256

Note: All values are measured in bits.

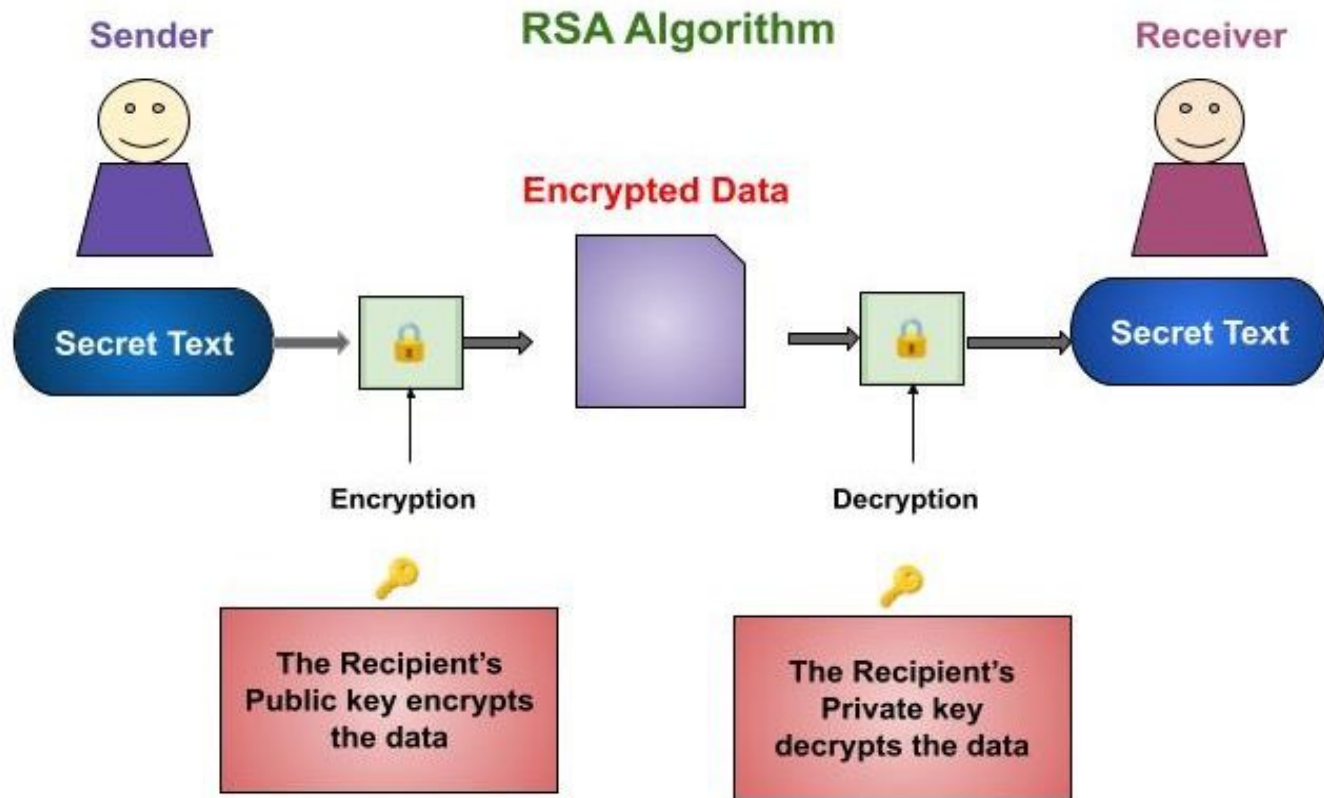
Public key Cryptography

- Public key cryptography is a method of encrypting or signing data with two different keys and making one of the keys, the public key, available for anyone to use.
- The other key is known as the private key.
- Data encrypted with the public key can only be decrypted with the private key.
- Because of this use of two keys instead of one, public key cryptography is also known as [asymmetric cryptography](#).

How does TLS/SSL use public key cryptography?

- Public key cryptography is extremely useful for establishing secure communications over the Internet (via HTTPS).
- A website's [SSL/TLS certificate](#), which is shared publicly, contains the public key, and the private key is installed on the [origin server](#) — it's "owned" by the website.
- [TLS handshakes](#) use public key cryptography to authenticate the identity of the origin server, and to exchange data that is used for generating the session keys.
- A key exchange algorithm, such as RSA or Diffie-Hellman, uses the public-private key pair to agree upon session keys, which are used for symmetric encryption once the handshake is complete.
- Clients and servers are able to agree upon new session keys for each communication session, so that bad actors are unable to decrypt communications even if they identify or steal one of the session keys from a previous session.

RSA Algorithm



- Let's say we choose $p = 61$ and $q = 53$.
- $n = p * q = 61 * 53 = 3233$
- $(n) = (p-1)(q-1) = 60 * 52 = 3120$
- Let's choose $e = 17$ (a common choice).
- $d = 2753$ (computed such that $(17 * d) \bmod 3120 = 1$)
- So, the public key is $(3233, 17)$, and the private key is $(3233, 2753)$.

- Encryption
- To encrypt a message, the sender Uses the public key (n, e) provided by the recipient. Converts the plaintext message (m) into a ciphertext (c) using the formula –
- $c = m^e \bmod n$.
- Suppose we want to encrypt the message "HELLO". We convert it to ASCII: H(72) E(69) L(76) L(76) O(79).

- We will encrypt each ASCII value separately –
- H: $C = 72^{17} \bmod 3233 = 2103$
- E: $C = 69^{17} \bmod 3233 = 2464$
- L: $C = 76^{17} \bmod 3233 = 2190$
- L: $C = 76^{17} \bmod 3233 = 2190$
- O: $C = 79^{17} \bmod 3233 = 875$

- Decryption
- Once the ciphertext (c) is received, the recipient uses their private key (n, d) to decrypt the message. Computes the plaintext message (m) using the formula –
- $m = c^d \bmod n$.
- The recipient receives the ciphertext (2103, 2464, 2190, 2190, 875).

- They use their private key to decrypt each value –
 - $M = 2103^{2753} \bmod 3233 = 72$ (H)
 - $M = 2464^{2753} \bmod 3233 = 69$ (E)
 - $M = 2190^{2753} \bmod 3233 = 76$ (L)
 - $M = 2190^{2753} \bmod 3233 = 76$ (L)
 - $M = 875^{2753} \bmod 3233 = 79$ (O)
- the decrypted message is "HELLO".

01

Step

Key Generation

Choose two prime numbers: $p = 3, q = 11$

Calculate $n = p * q = 33$

Calculate Euler's Totient Function: $\Phi(33) = \Phi(3) * \Phi(11) = 2 * 10 = 20$

Choose $e = 7$, which is co-prime with 20

Calculate d as the multiplicative inverse of e (7), so $d = 3$

Public Key = $(n, e) = (33, 7)$ Private Key = $(n, d) = (33, 3)$

RSA Encryption and Decryption Example

02
Step

Sharing of Public Key

Public Key = $(n, e) = (33, 7)$ Private Key = $(n, d) = (33, 3)$



Sender

$(n, e) = (33, 7)$

The Public Key is shared
with the Sender and the
Private Key is kept secret
with the Receiver .



Receiver

$(n, d) = (33, 3)$

RSA Encryption and Decryption Example

03
Step

Encryption - Message Conversion

Sender's Message(M) = "AC"



Numeric Conversion
(A - Z => 1 - 26)

13

Numeric Representation of "AC"

04
Step

Encryption Formula:

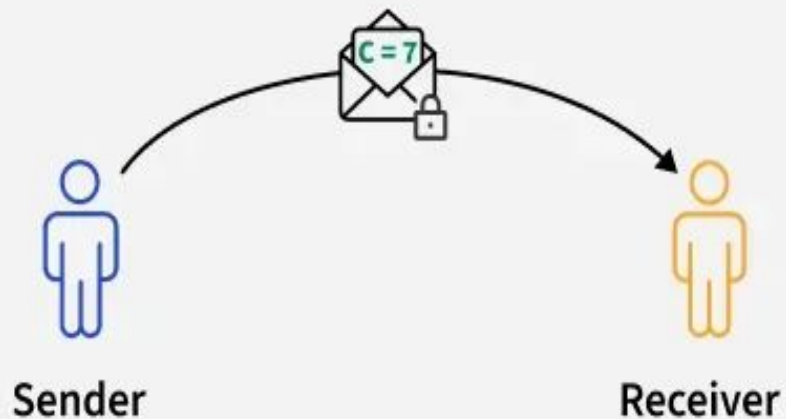
Encrypt the message using the Public Key (33, 7)

$$\text{Cipher Text } C = M^e \bmod n$$

$$C = 13^7 \bmod 33$$

$$C = 62748517 \bmod 33$$

$$C = 7$$



05
Step

Decryption Formula:

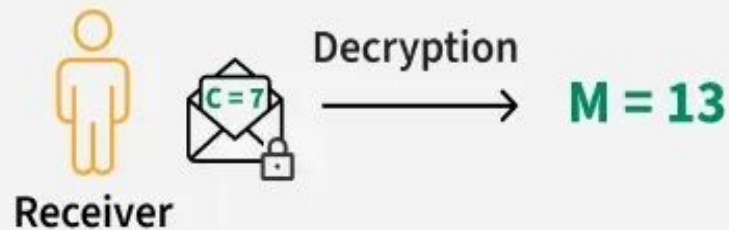
Decrypt the Cipher text using the Private Key (33, 3)

$$\text{Decrypted Text } M = C^d \bmod n$$

$$M = 7^3 \bmod 33$$

$$M = 343 \bmod 33$$

$$M = 13$$



The receiver uses decrypted text $M = 13$ to get the original message = "AC".

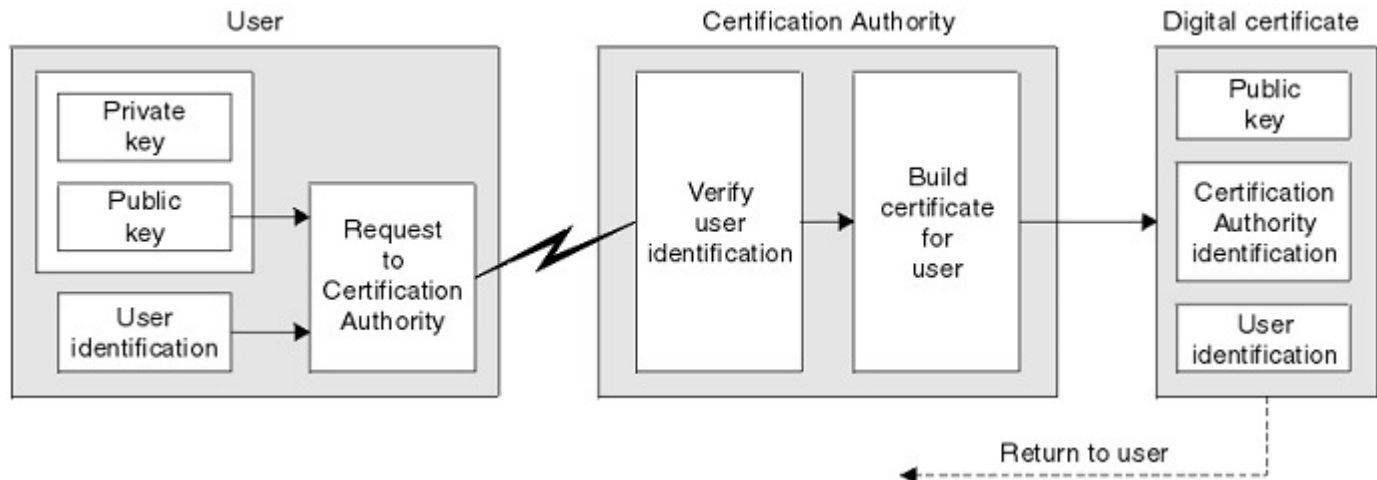
Digital Certificate

- A Digital Certificate referred to as a public key certificate, is used to establish a cryptographic link between the owner of a public key and that entity. These are used to exchange public keys needed for authentication and encryption.
- The certified public key, metadata about the certificate holder, identity information about the public key, and a **digital signature** of the public key created by the certificate issuer make up a **digital certificate**.

- It is basically a certificate issued digitally, issued to verify a user's authenticity i.e., verifying the user sending a message is who he or she claims to be, and also to provide the receiver with the means to encode a reply.
- Whoever wants to or an individual who wants to send encrypted messages applies for a digital certificate from a Certificate Authority (CA).

Need of digital certificate

- The digital certificate allows entities to share their public key in an authenticated way.
- They are used in initializing and establishing secure SSL (Secure Sockets Layer) connections between web browsers and web servers.



Digital Signature

- A digital signature or a digital code which is generated and authenticated by public key encryption is a mathematical scheme verifying the authenticity of digital messages or documents.
- The digital signature is based on RSA and public key cryptography which depends on two mutually authenticating cryptographic keys.
- The creator and the sender creating the digital signature uses his/her own private key to encrypt signature-related data and to decrypt that data the signer's public key is used which is the only key to decrypt it.

Purpose of digital signature

- It is attached to an electronically transmitted document verifying its contents and the sender's identity authenticating him.
- It is used for software distribution, financial transactions, and contract management software.

Public Key Infrastructure (PKI)

- Public Key Infrastructure (PKI) is a security framework that issues and manages digital certificates to establish trust on digital networks. It ensures secure communication by validating identities, protecting data, and preventing unauthorized access.
- Provides unique digital identities to users, devices, and systems
- Secures communication using public and private key pairs
- Prevents MITM attacks by verifying key ownership
- Ensures confidentiality, integrity, and authenticity of data